

UNIVERSIDADE DE LISBOA
FACULDADE DE CIÊNCIAS
DEPARTAMENTO DE BIOLOGIA



Functional characterization of the neural circuits underlying the optokinetic reflex in larval zebrafish

Manuel Pinho Gonçalves Bento de Bettencourt

Mestrado em Biologia do Organismo e Evolução

Versão Provisória

Dissertação orientada por:
Doutora Claudia Feierstein
Professora Doutora Clara Amorim

Acknowledgments

I would like to thank everyone who has contributed, academically or technically, to the successful completion of this thesis. This includes:

My colleagues at the ‘Vision to Action’ lab, namely Aaron, Catarina, Inês, Joaquim, Lucas Mariana and Pedro, who assisted or supported me at different times.

Claudia, Ruth and Michael, who supervised and advised my work;

The Champalimaud foundation’s scientific and technological platforms, especially the Fish Facility, which sustains zebrafish research, and the imaging platform, which provided help regarding confocal imaging;

The Fundação para a Ciência e a Tecnologia, which provided the funding.

I would also like to thank those who supported me on a more personal level, especially Carolina, my parents, and my sister, and to express my profound gratitude to my grandmother and Bruno, who have supported me throughout my academic journey.

Declaration on the use of Artificial Intelligence (AI)

The Large Language Models ChatGPT, Gemini and Claude were used to develop and/or debug the Python and MatLab scripts used in this work. The AI tool SciSpace was used to perform bibliographic searches. No generative AI was used to conceive, organize, or to produce any image or text in this thesis.

Abstract

Understanding how the brain interprets sensory information and conveys it into appropriate motor behaviour is a fundamental question in neurobiology. Simple, reflexive eye movements are an established model to study these sensory-motor transformations. One such reflexive behaviour is the optokinetic reflex, an innate response to whole-field visual motion in which the eyes move smoothly to track the rotation of the visual field. In this work, the neural basis of the optokinetic reflex in larval zebrafish (*Danio rerio*) was explored. Transgenic larvae expressing a genetically encoded calcium indicator pan-neuronally and a fluorophore in a specific subset of neurons (glutamatergic or GABAergic) were imaged with a two-photon microscope while exposed to optokinetic stimuli. By then relating neural activity with oculomotor variables, neurons with activity related to eye position and eye velocity were identified. After analysing their distribution, it was found that they are spatially segregated, forming distinct clusters. Finally, the neurotransmitter composition of these clusters was assessed, contributing to their characterization. The fidelity of the glutamatergic line as a reporter of neurotransmitter identity was also quantitatively evaluated, revealing a modest labelling accuracy. The findings presented here replicate and expand previous knowledge regarding the neural populations involved in the optokinetic reflex, and help establish methods for the analysis of neural circuits.

Keywords: optokinetic reflex; oculomotor system; functional imaging; larval zebrafish

Resumo

Um dos objetivos fundamentais da neurobiologia é compreender como é que o sistema nervoso é capaz de transformar as informações do mundo exterior, provenientes dos sentidos, em comandos motores capazes de desencadear uma ação. Estas transformações, denominadas transformações sensoriais-motoras, estão por detrás de todos os comportamentos, desde o mais simples reflexo até ao mais elaborado movimento voluntário, mas pouco se sabe sobre como acontecem.

Um conjunto de reflexos - os movimentos oculares reflexivos - tem servido nas últimas décadas como paradigma experimental para o estudo destas transformações. Estes movimentos são fundamentais para a nossa capacidade de ver, uma vez que permitem a estabilização do olhar aquando do movimento. Sem eles, começar a andar seria suficiente para perder toda a acuidade visual. Mas além da sua importância, os movimentos oculares estão presentes em todos os vertebrados e são comportamentos simples e estereotípicos, desempenhados por um sistema neuro-muscular também ele simples - o sistema oculomotor. Deste modo, são um bom modelo de estudo. Nesta tese, um destes reflexos, o reflexo optocinético, foi empregue como modelo.

O reflexo optocinético é uma resposta inata visual ao movimento rotativo (que acontece, por exemplo, quando se vira a cabeça) em que os olhos seguem o sentido de movimento da imagem visual. Desta forma, a imagem fica ancorada na retina, permitindo que seja percebida com detalhe. Trata-se de uma resposta estereotípica, em que os olhos se movem de maneira suave e contínua no sentido do movimento, mas quando chegam a posições limite são recentrados por movimentos balísticos chamados movimentos sacádicos. Experimentalmente, este reflexo pode ser facilmente evocado ao expor o animal a um padrão às riscas em constante rotação, que funciona como estímulo optocinético.

Nas últimas décadas, o estudo dos movimentos oculares em mamíferos permitiu esclarecer muitas questões em relação ao funcionamento do sistema oculomotor e dos diferentes movimentos oculares, incluindo o reflexo optocinético. Contudo, mais recentemente, um novo modelo animal, a larva de peixe zebra, tem ajudado a aprofundar este conhecimento.

O peixe zebra (*Danio rerio*) é um peixe de água doce amplamente utilizado em investigação biológica e biomédica, mas no seu estado larvar, com apenas alguns dias de idade, apresenta características particularmente apelativas para a investigação em neurobiologia. O seu cérebro é pequeno e transparente, o que possibilita olhar diretamente para o cérebro inteiro, ou uma grande parte dele, com métodos de microscopia especializados. Simultaneamente, trata-se de um vertebrado, cujo cérebro segue a mesma organização básica e contém estruturas homólogas ao cérebro humano. Por último, a larva apresenta já um variado repertório de comportamentos, incluindo o reflexo optocinético, que envolvem transformações sensoriais-motoras.

A investigação neurobiológica em larva de peixe-zebra baseia-se essencialmente na imagiologia funcional, uma abordagem que permite observar a atividade neuronal de um organismo vivo em tempo real. Este método baseia-se geralmente na utilização de peixes transgénicos que expressam um indicador de atividade, como por exemplo a molécula GCaMP, nos neurónios. Quando o neurónio é ativado, este indicador emite fluorescência, que pode depois ser captada com recurso a diversas técnicas de microscopia, incluindo a microscopia de dois-fótons. Assim, ao expor o peixe a determinado estímulo, registando a resposta comportamental do animal enquanto se grava a sua

atividade neuronal, torna-se possível relacionar diretamente a atividade neuronal com o comportamento.

Estudos prévios de imagiologia funcional em larva de peixe zebra, focados nos movimentos oculares, permitiram perceber melhor que áreas do cérebro estão envolvidas no controlo dos mesmos, como é que os circuitos neuronais processam o movimento visual rotativo, como é que os movimentos sacádicos são desencadeados, ou como a informação sobre a posição dos olhos é armazenada, entre outras questões. Contudo, ainda há muito por esclarecer. Em particular, o reflexo optocinético tem sido pouco estudado, especialmente no que concerne à componente contínua do movimento ocular neste reflexo. E apesar da imagiologia funcional ser um método muito poderoso, a compreensão dos circuitos neuronais exige mais informação, que pode provir de estudos de perda e ganho de função, de análises de conectividade neuronal, ou da identificação da identidade genotípica dos neurónios.

Neste contexto, este projeto surgiu com o intuito de aprofundar o conhecimento dos circuitos neuronais subjacentes ao reflexo optocinético. Para o fazer, pretendeu-se aliar uma experiência de imagiologia funcional, baseada num protocolo previamente estabelecido, com a identificação do tipo neuronal dos neurónios identificados (excitatório ou inibitório), baseado no neurotransmissor que produzem.

Em concreto, recorreu-se a imagiologia funcional do rombencéfalo (a parte posterior do cérebro) de larvas de peixe zebra transgénicas expostas a um estímulo optocinético, em que a atividade neuronal foi gravada com um microscópio de dois-fotões. Estas larvas exprimiam GCaMP no núcleo de quase todos os neurónios e um outro fluoróforo num subgrupo de neurónios baseado no tipo de neurotransmissor: ou nos neurónios glutamatérgicos, que produzem glutamato - o principal neurotransmissor excitatório - ou nos neurónios GABAérgicos, que produzem GABA - o principal neurotransmissor inibitório.

A identificação do tipo neuronal pressupõe que a linha transgénica utilizada marca corretamente os neurónios pretendidos. Para avaliar este pressuposto, realizou-se uma experiência de validação da linha transgénica *vglut2a:DsRed*, que marca neurónios putativamente glutamatérgicos. A validação baseou-se na realização de uma hibridização *in-situ* com sondas para os transcritos *vglut2a* e *vglut2b*, e na comparação da marcação resultante com a expressão observada na linha transgénica de uma forma quantitativa. Chegou-se à conclusão que a linha tinha uma precisão de 74% e uma sensibilidade de 62%, o que levanta algumas preocupações em relação ao seu uso.

Os dados de imagiologia funcional foram processados utilizando um pipeline de processamento em Python, permitindo extrair a atividade de cada neurónio, que foi depois correlacionada com uma série de variáveis relacionadas com o comportamento e o estímulo apresentado.

Esta análise revelou uma distribuição ampla e organizada de neurónios cuja atividade se relaciona com a posição ou velocidade ocular, em concordância com estudos anteriores. Uma análise subsequente baseada em regressão linear permitiu estabelecer a existência de várias populações neuronais relacionadas diferentemente com as duas variáveis. Algumas destas populações correspondem a núcleos com funções conhecidas, nomeadamente o núcleo *abducens* e o núcleo olivar inferior, mas não todas. Por último, a composição destas populações em termos de neurónios excitatórios e inibitórios foi determinada.

Quais são as implicações dos resultados obtidos? A experiência de validação da linha transgénica levanta preocupações em relação ao uso de linhas transgénicas como fonte de informação sobre a identidade neuronal. Tendo em conta a utilização generalizada destas mesmas na investigação atual, estes resultados são sobremaneira relevantes.

Quanto às experiências de imagiologia funcional, permitiram essencialmente enquadrar e complementar investigação anterior. Várias observações relativas à distribuição espacial de neurónios relacionados com a posição ou a velocidade ocular são concordantes com estudos anteriores. O foco no rombencéfalo inteiro permitiu ainda contextualizar resultados de outros artigos com áreas de foco mais restritas. Contudo, várias questões relativas aos circuitos implicados no reflexo optocinético ficaram por esclarecer, nomeadamente em relação aos mecanismos de integração e armazenamento de informação relacionada com a velocidade e posição ocular. Quanto à identificação de neurónios excitatórios e inibitórios, a determinação da composição das populações identificadas em termos de neurotransmissores foi pouco conclusiva, mas serve de exemplo ilustrativo do tipo de informação que se pode obter com a abordagem experimental usada.

Em suma, os resultados desta tese vão de encontro a, e aprofundam, o conhecimento prévio sobre as populações neuronais subjacentes ao reflexo optocinético. Além disso, contribuem para estabelecer um método para a análise dos circuitos neuronais baseado na imagiologia funcional e na identificação do tipo neuronal.

Palavras-chave: Reflexo optocinético, peixe-zebra, sistema oculomotor, imagiologia funcional

Index

Acknowledgements	ii
Declaration on the use of artificial intelligence (AI)	ii
Abstract	iii
Resumo	iv
Index	vii
List of figures and tables	x
List of abbreviations and acronyms	xii
1. Introduction	14
1.1. Overview	14
1.2. Thesis outline	15
2. Background	16
2.1. The oculomotor system and the control of eye movements	16
2.1.1. Eye movements and the oculomotor system	16
2.1.2. Principles of oculomotor control	18
2.2. Larval zebrafish as a model in neurobiology	20
2.2.1. Features and advantages of larval zebrafish for neurobiology	20
2.2.2. Functional imaging in larval zebrafish neurobiology research	22
2.3. Progress in oculomotor research with larval zebrafish	23
2.3.1. The neural basis of integration	23
2.3.2. The control of saccades	24
2.3.3. Responses to optic flow: the OMR and the OKR	25
2.4. Conclusion	25
3. Methods	27
3.1. Line validation	27
3.1.1. <i>In-situ</i> hybridization	27

3.1.2. Confocal image acquisition and processing	27
3.2. Imaging neural activity during behaviour	28
3.2.1. Two-photon imaging and simultaneous behaviour recording	28
3.2.2. Behaviour data processing	29
3.2.3. Functional image processing	29
3.3. Data analysis of functional imaging experiments	30
3.3.1. Identification of neuronal neurotransmitter identity	30
3.3.2. Building an anatomical reference brain	30
3.3.3. Correlation Analysis	30
3.3.4. Eye position and eye velocity tuning	31
4. Results	33
4.1. Fidelity of the transgenic vglut2a:DsRed line	33
4.1.1. Qualitative overview of line fidelity	33
4.1.2. Quantitative characterization of line fidelity	34
4.2. Neural correlates of oculomotor behaviour	35
4.2.1. Correlations between neural activity and oculomotor variables	36
4.2.2. Spatial distribution of oculomotor-tuned neurons	37
4.2.3. Eye position and velocity encoding regions	40
4.2.4. Neurotransmitter identity of outlined eye position and velocity clusters	43
5. Discussion	44
5.1. Overview	44
5.2. Validating the transgenic lines used in this work	44
5.3. Oculomotor related activity in the hindbrain	45
5.3.1. Eye position and eye velocity coding in the larval zebrafish hindbrain	45
5.3.2. Neural populations involved in the OKR	46
5.3.3. Location of the hVPNI and VSM	48
5.4. Conclusion	48
6. Bibliography	50
 Appendix A – Comparative neuroanatomy of eye movement control	 56
A.1. General anatomy of the brainstem	56
A.2. Anatomical localization of oculomotor control structures in mammals	56

A.3 Anatomical localization of oculomotor control structures in fish	57
Appendix B – Zebrafish strains and animal care	59
Appendix C – Confocal image processing	60
Appendix D – Functional image processing	61
D.1 Motion correction	61
D.2 Nuclei segmentation	61
D.3 Training a StarDist2D model	62
Appendix E – Building an anatomical reference brain	65
Appendix F – Supplementary figures	66

List of figures and tables

Figures

Figure 2.1 - The Optokinetic Reflex.	17
Figure 2.2 - Diagram of the oculomotor system.	18
Figure 2.3 - Pre-oculomotor control.	20
Figure 2.4 - The larval zebrafish brain.	21
Figure 2.5 - Functional imaging using genetically encoded GCaMP and two-photon imaging.	23
Figure 3.1 - Schematic of the python pipeline developed in this work to process functional imaging datasets.	29
Figure 4.1 - A representative example of a confocal imaging experiment.	33
Figure 4.2 - ISH versus DsRed labelling.	34
Figure 4.3 - A representative example of a functional imaging experiment.	36
Figure 4.4 - Observed correlations are significant.	37
Figure 4.5 - Eye position correlation maps.	38
Figure 4.6 - Slow eye velocity correlation maps.	39
Figure 4.7 - Position-velocity index map.	41
Figure 4.8 - Manually outlined neuron clusters.	42
Figure 4.9 - Neurotransmitter composition of the neural clusters.	45
Figure D.1 - StarDist model training.	64
Figure E.1 - Determining ABN and OMN location with the chata:mCherry line.	65
Figure F.1 - Performance of the automatic classification of neurons.	66
Figure F.2 - Gad1b fish had less segmented neurons than vglut2a fish.	67
Figure F.3 - Observed correlations are significant (data for all fish).	67
Figure F.4 - Nasal-temporal position distinction.	68
Figure F.5 - Classification of neurons from 2-photon imaging stacks as DsRed+ and DsRed-.	68

Tables

Table 3.1 - Regressors used in the correlation analysis.	32
Table 4.1 - Results from the quantitative validation of the transgenic line.	35

Table B.1 - Transgenic Lines used in this work.	51
--	----

List of abbreviations and acronyms

A – Anterior

ABN – Abducens Nucleus

AI – Artificial Intelligence

ANTs – Advanced Normalization Tools

AOS – Accessory Optic System

APT – Area Pretectalis

BAC – Bacterial Artificial Chromosome

BN – Burst Neurons

CDpos – Caudal Dorsal Position Nucleus

CLAHE – Contrast Limited Adaptive Histogram Equalization

CLmix – Caudal Lateral Mixed Nucleus

CMTK – Computational Morphometry Toolkit

D – Dorsal

dpf – days post-fertilization

DTN – Dorsal Terminal Nucleus

EV – Explained Variance

FM – Full Model

FN – False Negative

FP – False Positive

GABA – Gamma-Aminobutyric Acid

HBO – Hindbrain Oscillator

HCR – Hybridization Chain Reaction

hVPNI – horizontal Velocity-to-Position Neural Integrator

IO – Inferior Olive

IoU – Intersection over Union

ISH – *In-Situ* Hybridization

L – Left

LR – Lateral Rectus

LSM – Light Sheet Microscope

MeOH – Methanol

MHbPos – Middle Hindbrain Position Nucleus

mitfa – melanocyte inducing transcription factor A

MLF – Medial Longitudinal Fasciculus

MR – Medial Rectus

NOT – Nucleus of the Optic Tract

NPH – Nucleus Prepositus Hypoglossi

OKAN – Optokinetic After Nystagmus

OKR – Optokinetic Reflex

OMN – Oculomotor Nucleus

OMR – Optomotor Response

P – Posterior

PBS – Phosphate Buffered Saline

PBST – Phosphate Buffered Saline Tween

PFA – Paraformaldehyde

PMT – Photomultiplier

PPRF – Paramedian Pontine Reticular Formation

PVI – Position-Velocity Index

R – Right

RM – Reduced Model

ROI – Region of Interest

STD – Standard Deviation

TN – True Negative

TP – True Positive

V – Ventral

VN – Vestibular Nuclei

VOR – Vestibular-Ocular Reflex

VPNI – Velocity-to-Position Neural Integrator

VSM – Velocity Storage Mechanism

1. Introduction

1.1. Overview

Understanding the neural basis of sensory-motor transformations is a major research question in neurobiology. One of the most striking features of animal behaviour is the ability to flexibly engage different actions or movements in response to different stimuli from the outside world. For this to be possible, information about the outside world must be sensed and decoded through sensory systems. Then, at the end of the process, precise and coordinated commands need to be sent to the muscles, instructing them how to perform the desired action. Both sensory transduction and motor control are highly fascinating processes which have been the focus of much research in neuroscience. But in between them there needs to occur a *sensory-motor* transformation: from the sensory evidence (what is going on) to the motor action (what to do). In fact, from the simplest reflexive behaviours to the more elaborate voluntary movements, sensory-motor transformations are constantly playing out in the brain. Nevertheless, neuroscience is still far from understanding how these transformations happen.

The study of eye movements has served as a powerful model to explore sensory-motor transformations¹. Although often unnoticed, eye movements are essential for our ability to see, mainly due to gaze stabilization²: Anyone who has tried a camera should know how difficult it is to maintain a focused and stabilized image while moving, and that is a problem our eyes and brains must deal with as well. Indeed, eye movements evolved to respond to this very fundamental problem in animals: moving and seeing simultaneously. That is why they are conserved throughout the vertebrate lineage, from lampreys to humans². Such conservation, along with their simplicity and easy experimental accessibility, makes eye movements an excellent model to understand how sensory information is used to control motor action.

Throughout the last decades, studies in mammals were essential in unravelling the inner workings of the oculomotor system, the neuro-muscular system responsible for eye movements^{3,4}. More recently, an emerging model in neuroscience, larval zebrafish (*Danio rerio*), has taken centre stage in this challenge. The small size and optical accessibility of the larval zebrafish brain make it possible to non-invasively record the neural activity of the whole brain during behaviour, something currently not possible in classical vertebrate models⁵. Moreover, larval zebrafish present a diverse repertoire of visuo-motor behaviours⁶ which can be easily elicited with visual stimulation and measured with high-speed tracking cameras. Accordingly, there is a growing body of research that has focused on using neural activity imaging (commonly called functional imaging) of the larval zebrafish brain to study the neural correlates of visual-to-motor transformations⁷.

The work done for this thesis follows the aforementioned research line, focusing specifically on studying the neural correlates of the optokinetic reflex (OKR) in the hindbrain region. The OKR is a response to rotational optic flow (i.e. movement of the visual field), characterized by a smooth compensatory movement of the eyes interrupted by fast resetting movements¹. Previous research has contributed to dissecting what areas of the brain are involved in this behaviour, how optic flow is represented and processed in the brain, how eye movements are controlled and how stimulus velocity is converted into desired eye position⁸⁻¹². Yet a deep understanding of the neural circuits underlying this reflex is far from achieved. Namely, it is not clear how eye position and velocity information is stored and used to control slow eye movements. And although functional imaging is a very powerful method, incorporating new layers of information beyond the neural activity data is crucial to be able to fully understand neural circuits. One of these layers is determining the neurotransmitter identity of imaged neurons - and thereby seeing if they are excitatory or inhibitory - which opens the door to

more realistic and detailed models of circuit function¹². Different approaches can be employed to retrieve this information. In the work presented here, by leveraging on transgenic fish lines, for each imaged fish it was possible to both record the neural activity of the neurons within the hindbrain and check for a neurotransmitter-specific labelling.

The key purpose of the thesis project was therefore to further the understanding of the neural circuits underlying oculomotor behaviour, and more specifically, the OKR, by determining the excitatory and inhibitory nature of the neuronal populations active during this behaviour.

Within this general purpose, the following was achieved:

- Through a validation experiment following previously developed protocols, a quantitative assessment of the transgenic line targeting glutamatergic neurons was obtained;
- A python pipeline relying on open-source tools was developed to process the functional imaging data;
- Imaging experiments of several fish with glutamatergic (excitatory) or GABAergic (inhibitory) labelling were performed using a previously developed experimental paradigm;
- Following a regression analysis of the imaging experiments, which related neural activity to recorded oculomotor behaviour, eye-position and eye-velocity related clusters of neurons in the hindbrain were outlined;
- The composition of the identified neural clusters in terms of putatively excitatory or inhibitory neurons was determined, which could contribute to understand better the neural populations underlying the OKR.

1.2. Thesis Outline

The thesis is structured in five main chapters:

In **chapter 1**, the overarching scientific goal of the project is discussed, as well as a brief presentation of the main results.

In **chapter 2**, the theoretical background of the work is presented with a focus on the oculomotor system and the control of eye movements, the use of larval zebrafish as a model in neurobiology and a review of recent research using larval zebrafish to investigate the oculomotor system.

In **chapter 3**, the methodology of the experimental and computational work is presented.

In **chapter 4**, the results of the project are presented, including the assessment of the transgenic line and the data analysis from functional imaging experiments.

In **chapter 5**, the results of the work are discussed.

Further supporting information can be found in the Appendices.

2. Background

The present chapter reviews the theoretical background of the research entailed in this thesis. It starts by explaining the basic knowledge about eye movements, which include the optokinetic reflex, and the oculomotor system underlying them. It also delves into some of the foundational research done with mammals that gave rise to the current understanding of the sensory-motor transformations and control mechanisms behind oculomotor behaviour. Afterwards, the larval zebrafish model is introduced, and there is a brief presentation of its relevant characteristics regarding neurobiology. The principle behind functional imaging is also explained. Finally, in the last part of the chapter, a brief review of recent literature focusing on understanding larval zebrafish oculomotor behaviour is undertaken. A discussion on the neuroanatomy of the different elements presented throughout this chapter can be found in Appendix A.

2.1. The oculomotor system and the control of eye movements

The study of eye movements has served as a paradigm in sensory-motor research for the last decades^{1,13,14}. These behaviours are underpinned by a simple neuromuscular system, the oculomotor system, which involves a remarkably elaborate control¹. Modelling approaches and experimental studies in mammals were essential in unravelling the inner workings of this system¹⁵⁻¹⁷, contributing to the bulk of the understanding regarding eye movement control that exists today. In this section, the different types of eye movements and the main control elements of the oculomotor system are discussed.

2.1.1. Eye movements and the oculomotor system

There are three main eye movement types involved in gaze stabilization in vertebrates: the Vestibular-Ocular Reflex (VOR), the Optokinetic Reflex (OKR) and Saccadic eye movements. There are also two more derived eye movement types, Vergence and Smooth Pursuit, which will not be discussed further in this chapter. All of these movements are however based on the same repertoire of movements of the eyeball, which consists of fixations, i.e., the eye stays still; saccades, which are very fast and abrupt movements in one direction; and smooth, slow movements, in which the eye smoothly turns towards one direction¹.

The VOR is the basic gaze-stabilizing reflex during rotations of the head: signals from the vestibular organs detecting a turning movement promote a compensatory movement of the eyes in the opposite direction. The OKR has roughly the same function, but instead of using vestibular clues, it uses visual ones: when the head/animal turns in one direction, the whole visual image appears to move in the opposite direction. This motion, called retinal slip, triggers the OKR, in which the eyes follow the visual image, providing the needed compensatory movement. By doing so, the visual image is effectively anchored to the retina for a longer time, allowing for high-acuity vision².

These reflexes are quite similar: they are often engaged in the same situations and have similar kinematics. In both cases, the eyes accompany the visual image with smooth movements in the

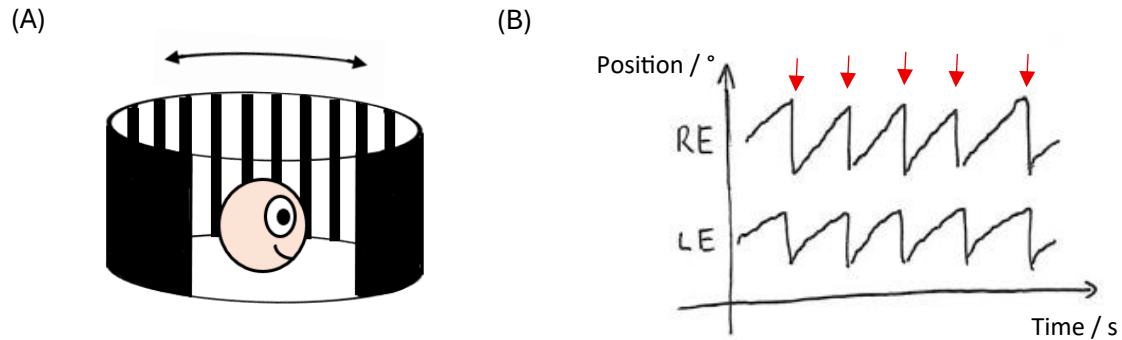


Figure 2.1. The Optokinetic Reflex. (A) Experimental paradigm to study the OKR. (B) Eye position plots during optokinetic stimulation, showcasing the slow eye movements and the saccades (red arrows).

direction of the visual flow/opposite direction of head rotation, but when they begin reaching their limit of motion, a saccade in the opposite direction resets the movement. This can be easily seen experimentally: the OKR can be triggered by presenting the animal with a rotating drum with black and white stripes. While there is no self-movement, the animal senses the whole-field rotation of the visual image, which gives the necessary retinal slip signal to activate the OKR. If the position of the eyes is recorded during such an experiment, both components of this reflex (i.e., saccades and smooth movement) can be seen, as shown in Figure 2.1.

Saccadic eye movements are performed during the OKR and VOR, to recentre eye position, but there are also spontaneous saccades. Their fundamental function is to allow for fixations in which the visual scene can be perceived². A similar behaviour is seen in birds: while a pigeon is walking forward, its head does bobbing movements. These allow the pigeon to fixate the visual image for longer, but then it must catch up with the movement of the body, which requires a quick readjustment of the position of the head, much like the saccade does for the eye¹⁸.

Executing this set of movements is the task of the oculomotor system, a simple neuromuscular system which consists of three pairs of extraocular muscles innervated by cranial nerves that originate in three different brainstem nuclei¹, as will be explored below. The focus will be on the horizontal oculomotor system, since it has been the focus of more research.

The horizontal movement of the eyes is controlled by an opposing pair of extraocular muscles - the Medial Rectus (MR) and Lateral Rectus (LR). When the MR contracts, the eye moves nasally, i.e., towards the nose. On the other hand, when the LR contracts, the eye moves temporally, i.e., away from the nose. The medial rectus is controlled by the ipsilateral (i.e. of the same hemisphere) oculomotor nucleus (OMN), to which it is connected by cranial nerve III (the oculomotor nerve), whereas the lateral rectus is controlled by the ipsilateral abducens nucleus (ABN), to which it is connected by cranial nerve VI (the abducens nerve). The ABN is also connected to the contralateral (i.e. of the opposite hemisphere) OMN via interneurons that travel along and across the midline in the medial longitudinal fasciculus (MLF).

To understand how the basic circuit underlying conjugate horizontal movements works, consider the case where the eyes move to the left (Figure 2.2). In this situation, the left eye must move temporally, while the right eye must move nasally. To accomplish this, a signal must first reach the left ABN. There, it will activate the lateral rectus motor neurons and the ABN interneurons. The lateral rectus motor neurons signal to the lateral rectus of the ipsilateral (left) eye to contract, leading to a temporal movement of this eye. Simultaneously, the interneurons cross the midline and relay their signal to the medial rectus motor neurons that are found in the contralateral OMN, signalling the

medial rectus of the right eye to contract, leading to a nasal movement of this eye. Thus, both eyes move co-ordinately to the left¹.

The basic architecture of the oculomotor system is well established, but it opens doors to new questions: What is the provenance and contents of the command signal that reaches the ABN? And how are different kinds of movement encoded? - In other words, how is this circuit controlled?

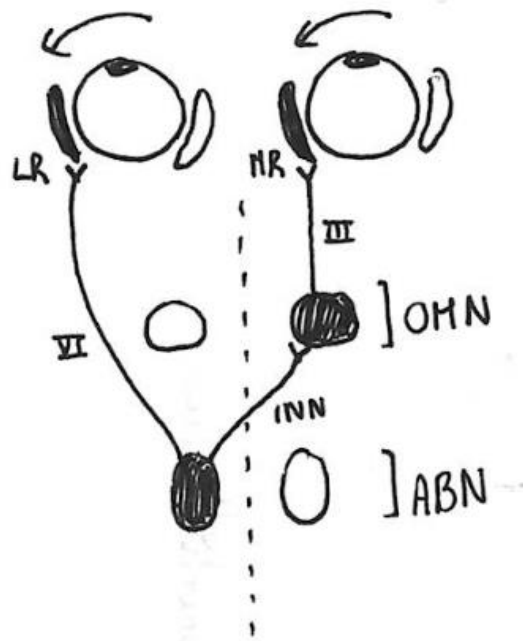


Figure 2.2. Diagram of the oculomotor system. Circuit for horizontal conjugate eye movements to the left. See text for details.

2.1.2. Principles of oculomotor control

A key observation coming from electrophysiological studies in primates is that motoneurons' activity depends on both the position and velocity of the eye¹⁵. This dependence necessarily relates to the mechanical properties of the eye muscles and other tissues in the orbit, collectively denoted as the eye plant. Indeed, consider the example of a spontaneous saccade followed by a fixation: for the eye to start moving, it needs a pulse of muscle contraction, proportional to eye velocity, in order to overcome the inertia of the plant. However, after the saccade, to maintain the eye still in an eccentric position, there needs to be an additional signal, proportional to eye position, that counteracts the natural tendency of the plant to return to its centred position.

Taking into account what was stated above, and the findings that in both the VOR and the OKR the input signal is a velocity signal¹⁹⁻²¹, it becomes apparent that there must be a mechanism that is able to convert these velocity inputs into a position signal that is required for the motoneurons. The transformation from velocity to position is easily obtained by performing integration - in the mathematical sense - and this is the role of a set of neurons known as the Velocity-to-Position Neural Integrator (VPNI)^{22,23}.

The importance of the VPNI was clearly demonstrated by lesion studies that targeted this structure^{17,24}: after the lesion, the animals were incapable of maintaining eye fixations following saccades (Figure 2.3A). Understanding the role that velocity integration has in slow eye movements is more challenging, however. Cannon and Robinson¹⁷ suggest that without the VPNI, the only signal reaching motoneurons during the slow component of the VOR is a constant velocity signal (a step), which leads to a step-change in eye position. With the VPNI, in addition to this signal there is a ramp signal resulting from integration of the velocity step, which leads to a constant displacement of the eye, i.e., a step-change in eye velocity.

Detailed studies of the kinematics of eye movements have also unravelled a separate velocity integration process in the oculomotor system. Cohen and colleagues¹⁶ looked at slow eye velocity during the OKR and an associated after-effect, the Optokinetic After Nystagmus (OKAN), in which the eyes continue to rotate in the absence of any stimuli. They saw that during the OKR, the slow velocity is not constant but rapidly rises at first and then builds up to the maximum eye velocity, and during OKAN, there is an exponential drop (Figure 2.3B). This led to the proposal of a Velocity Storage Mechanism (VSM), a separate integrator which can store and accumulate velocity signals, and in the absence of further input, slowly discharge them, leading to the OKAN. The function of such a mechanism remains unknown, but it has been proposed that it serves to provide a more accurate velocity signal, by integrating gravitational information²⁵, buffering abrupt changes in vestibular or optokinetic input²⁶, or by countering the physical limitations of the vestibular system²⁷.

In a simplified way, it could be said that these two elements, the VSM and the VPNI, participate in two of the key sensory-motor transformations occurring during eye movements: On the one hand, it appears that during the VOR and the OKR, the VSM takes part in the conversion of direct sensory information about head and environment velocity into desired eye velocity²⁵⁻²⁷. On the other hand, the VPNI plays a key role in transforming the desired eye velocity into the motor signal needed for the eyes to move with such velocity, by providing the motoneurons with the eye position signal they require.

To summarise, Figure 2.3C contains a schematic diagram of how the oculomotor system is thought to operate. There are at least three pathways, the vestibular, optokinetic and saccadic pathways. The specialized vergence movements and smooth tracking are not considered here. Information about head velocity comes to the vestibular and optokinetic pathways, creating a velocity signal. This signal is fed directly into the motoneurons, and to the VSM, where it is accumulated, but it is also integrated by the VPNI, resulting in a position command that is relayed to the motoneurons as well. The saccadic pathway is activated in several different contexts including during the OKR and VOR, or spontaneously. It directly gives a velocity signal to the motoneurons and to the VPNI, which integrates it and provides the needed position command^{16,17}.

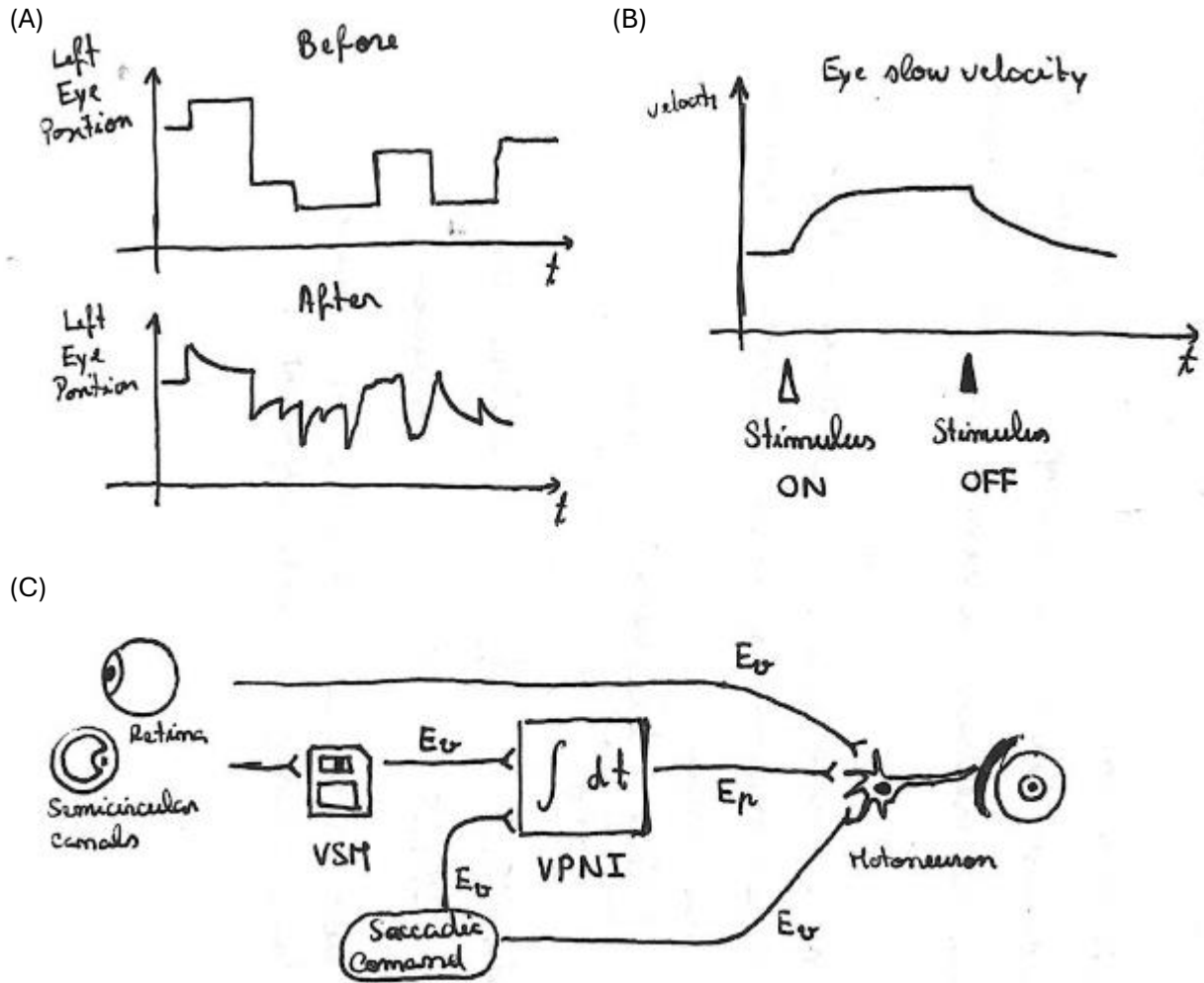


Figure 2.3. Pre-oculomotor control. (A) Results demonstrating the importance of the VPNI for maintaining fixations. Adapted from¹⁶. (B) Slow eye velocity (with saccades removed) during the OKR and OKAN suggested a second integrator, the VSM²⁴. (C) Simplified schematic of the pre-oculomotor system for the saccadic, optokinetic and vestibular eye movement pathways.

2.2. Larval zebrafish as a model in neurobiology

Zebrafish (*Danio rerio*) is a teleost fresh-water fish native to southeastern Asia²⁸. In the wild, it has been found mostly in shallow and still waters such as those from rice paddies²⁹. It is a popular aquarium fish and, since the 1970s, it has been a prominent model in experimental biology. Today, it has become a consolidated model in biological and biomedical research, in diverse areas from oncology to regenerative biology, including neuroscience^{30,31}. In this section this model is presented along with the main method employed to study it in neurobiology.

2.2.1. Features and advantages of larval zebrafish for neurobiology

For neuroscience, one of the biggest advantages of the zebrafish model is the optical accessibility of the brain in larval stages³². The larval cranium is transparent, and although the larvae are naturally pigmented, there are unpigmented mutants readily available³³. Moreover, in this stage,

the brain is quite small, occupying less than a cubic millimetre, with just 100000 neurons³⁴. Taken together, these two features mean it becomes possible to use sophisticated non-invasive imaging techniques to image large parts or even the whole brain of a vertebrate organism³⁵.

Furthermore, the zebrafish brain follows the same vertebrate blueprint of the human brain, divided into three compartments: the forebrain (prosencephalon), the midbrain (mesencephalon) and the hindbrain (rhombencephalon), and homologs to the main brain regions can be found not only in the adult but, even if rudimentary, in the larva at just five days of age³⁶ (Figure 2.4). Regarding neurochemistry, the zebrafish brain is also very similar to the mammalian one. Glutamate is the main excitatory neurotransmitter and gamma-aminobutyric acid (GABA) and glycine are the main inhibitory ones. Acetylcholine is the neurotransmitter of the neuromuscular junction and there are several neuromodulators, including serotonin, dopamine and noradrenaline. Glutamatergic and GABAergic neurons are widely distributed throughout the brain³⁷. Glycinergic neurons, on the other hand, are less widespread, but they are prominent in the hindbrain³⁸. Interestingly, in this region there is a conspicuous pattern in the organization of cells, with alternating parallel stripes of glutamatergic and glycinergic neurons³⁹.

For more fundamental neurobiology research, an additional advantage is that larval zebrafish have a diverse repertoire of sensory-driven behaviours that emerge from an early age (reviewed in detail by Fero and colleagues⁴⁰) and whose study has led to great advances in understanding sensory-motor transformations⁶. These behaviours include the startle response, hunting behaviour, the OKR and the optomotor response (OMR). Most of these are simple innate reflexes, but they already showcase complex sensory integration and sensory-motor transformations. When it comes to oculomotor behaviours, larval zebrafish do spontaneous and convergent saccades, the OKR, and, later in development, the VOR.

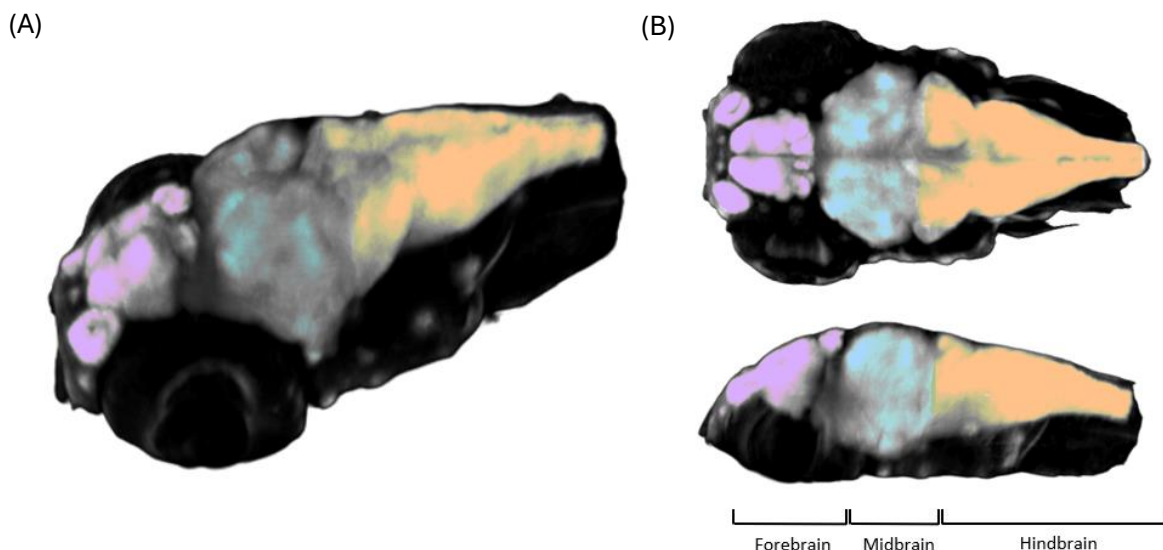


Figure 2.4. The larval zebrafish brain. Images taken from the Zebrafish Brain Browser⁴¹. (A) 3D view from the left rostral side. (B) horizontal (top) and sagittal (bottom) views of the larval brain. The main compartments are coloured: pink – forebrain, blue – midbrain, and orange – hindbrain.

2.2.2. Functional imaging in larval zebrafish neurobiology research

Given all the advantages presented above, a plethora of methods have been successfully established in larval zebrafish. These include molecular and genetic tools, behavioural recordings and advanced imaging techniques⁴². Of particular interest to neurobiology is the combination of behaviour recordings with live imaging of brain activity: By looking directly at the brain while the animal is behaving, the neural correlates of behaviour can be directly seen⁵. But how can this be achieved?

Imaging of live brain activity, commonly called functional imaging, is possible by using genetically engineered calcium activity indicators, such as GCaMP. When a neuron is activated, there is an influx of positive cations, including calcium, to the cytosol. This calcium binds calcium-binding sites within the GCaMP protein, promoting a conformational change that greatly increases its fluorescence (Figure 2.5A). By putting the expression of this protein under the control of a known promoter, it is possible to see the activity of most neurons in the brain (pan-neuronal expression) or of specific neuronal types.

Doing so requires using advanced fluorescence microscopy techniques such as two-photon microscopy, a variation of standard fluorescence microscopy. In the standard case, a fluorophore absorbs a photon, transitions to an excited state, and returns to the basal state, releasing a photon in the process (i.e., fluorescence). In the two-photon case, instead of absorbing one photon, the fluorophore absorbs two lower-energy photons simultaneously to get to the same excited state⁴³ (Figure 2.5B). To accomplish this, a laser must send a very concentrated pulse of photons of the right wavelength, to guarantee simultaneous absorption. Two advantages come from this system: on the one hand, there is less damage to the tissues and less photobleaching, allowing for longer acquisitions and imaging in live animals; on the other hand, background excitation is minimal, greatly improving the resolution of the images.

The great advantage of the functional imaging approach is that it offers the unparalleled opportunity to monitor neuronal activity across thousands of cells simultaneously, while knowing their spatial arrangement - a powerful method for understanding neuronal circuits⁵. Nonetheless, there are also drawbacks to this technique, namely, the diminished temporal resolution. Most GCaMP indicators have a relatively slow rise and decay in fluorescence, which obscures actual firing rates. In the end, functional imaging gives us a window into the activity of the brain *during* behaviour. But how can that data be used to understand the neural activity *underlying* behaviour? A common approach is to use regression, and other machine learning approaches, to relate neural activity (i.e. calcium activity) to a set of behaviour-related regressors. Often these regressors are first convolved with a GCaMP kernel that emulates the slow dynamics of the calcium sensor, a technique borrowed from functional magnetic resonance imaging⁴⁴. A neuron related to a particular behaviour can then be found by looking at the modelling parameters or correlation values (Figure 2.5C).

Notwithstanding the power of this approach, it can only find correlations, not causation. To probe further into the circuits, several other techniques are employed: gain and loss of function experiments can establish causality; labelling techniques can help understand neuronal identity and functional role within the circuit; and morphological tracing can verify the connectivity between neurons to further establish putative connections. As the next part of this chapter will show, most current research into zebrafish neurobiology uses a mixture of these methods to probe into how neural circuits work.

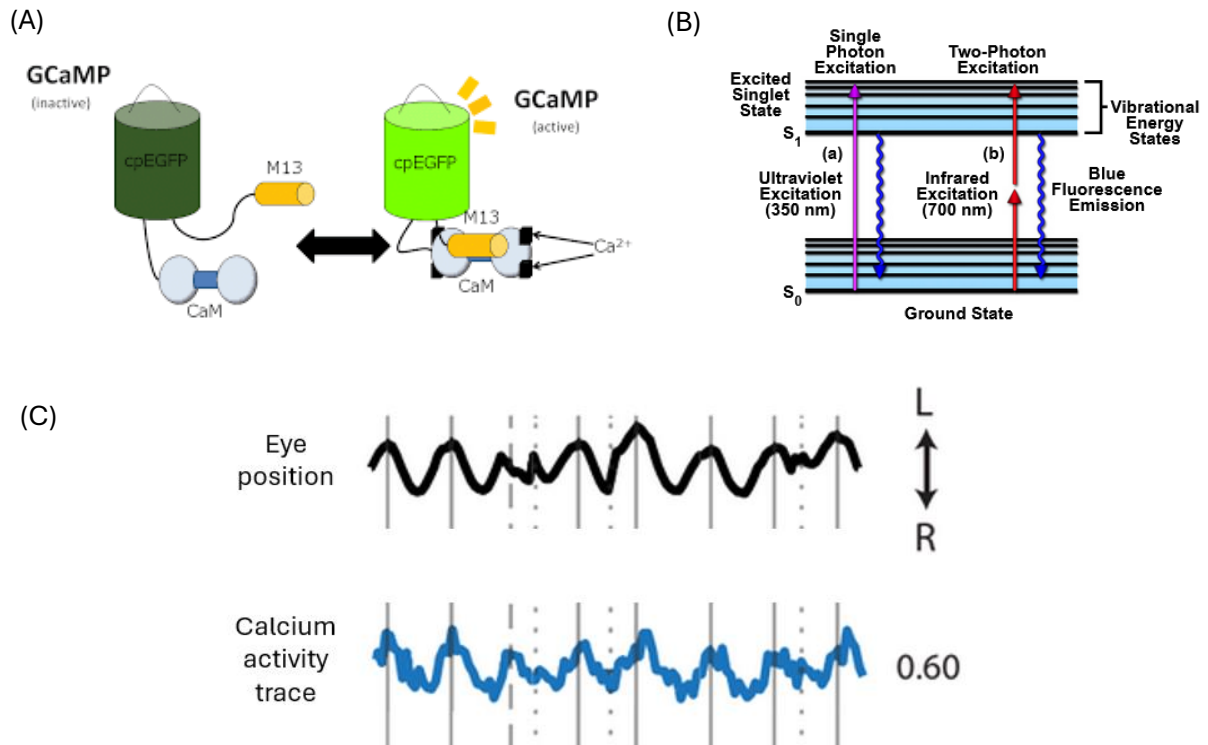


Figure 2.5. Functional imaging using genetically encoded GCaMP and 2-photon imaging. (A) GCaMP structure and mechanism. Image taken from⁴⁵. (B) Two-photon microscopy principle – taken from⁴⁶. (C) Finding behaviourally relevant neurons with regression and correlation. Plot adapted from¹². Top: eye position trace; Bottom: neuronal activity of one neuron with correlation with eye position equal to 0.60.

2.3. Progress in oculomotor research with larval zebrafish

The previous parts of this chapter presented two powerful models for the study of sensory-motor transformations: The vertebrate oculomotor system and larval zebrafish. It should therefore be clear that studying the oculomotor system in larval zebrafish is a promising approach to understand sensory-motor transformations. Indeed, there is a growing body of research that has focused on using functional imaging of larval zebrafish to study the neural correlates of visual-to-motor transformations, including oculomotor behaviour⁷. In the following text, some of these studies will be presented.

2.3.1. The neural basis of integration

The horizontal velocity-to-position integrator (hVPNI) has been a big focus of research within larval zebrafish neurobiology. This system is an interesting model for two main reasons. First, it performs mathematical integration. Second, it is a rudimentary form of short-term memory, as it can store eye position information for some time. Several authors have focused on how the cells in the hVPNI region could implement integration and memory storage. Leveraging on functional imaging¹¹, optogenetics⁴⁷, and machine learning approaches⁴⁸ these investigations showed a more complex picture than the traditional theoretical models had envisioned. Namely, Daie and colleagues⁴⁸ found

that the spatial pattern of neural activity within the hVPNI was context dependent, with different activity patterns during optokinetic and saccadic responses, leading them to propose a ‘multi-mode’ model.

Yet understanding how a circuit implements some computation depends not only on dissecting the activity patterns of cells within this circuit but also their nature, as excitatory or inhibitory. Lee and colleagues¹², profiting from the conspicuous hindbrain patterning of neurons according to neurotransmitter identity, looked at the neurotransmitter identity of the hVPNI neurons. They imaged transgenic fish expressing fluorophores in genetically defined neuronal classes, and identified the hVPNI neurons through functional imaging, discovering three main populations of neurons within the hVPNI: a central glutamatergic population with recurrent connections and connections to the ipsilateral ABN, in accordance with most theoretical models, and two other populations that connect to the contralateral side, one GABAergic and another glutamatergic, whose role is not entirely clear, suggesting nonetheless that there is bilateral coordination of neural integration.

2.3.2. The control of saccades

Saccades have long served as a model within the context of oculomotor research in primates⁴⁹. In zebrafish, several studies have also focused on understanding the control of saccades. Namely, there were two interesting studies, which will be explored below, that attempted to respond to two different questions: How is binocular control achieved during saccades? And how are different types of saccades encoded?

Leyden and colleagues⁵⁰ tackled the first question. The fact that most eye movements are conjugate, yet some are disconjugate, begets the question of how binocular control is achieved. This question has been a longstanding source of debate and two main theories have been proposed: Hering’s law of equal innervation postulated the eyes to receive commands from the same source, as ‘a pair of horses’ is controlled by a ‘single reins’⁵¹, whereas the Helmholtzian view proposed separate circuits controlling the different eyes, in which case binocular coordination would have to be learned⁵². Following on previous work⁵³, Leyden and colleagues⁵⁰ set out to find the monocular versus binocular nature of saccadic control. To do so, they presented an optokinetic stimulus designed to provoke more disconjugate and monocular saccades, with independent stimuli projected to both eyes. By then looking at neuronal calcium activity during behaviour, they were able to identify neurons that responded to specific types of saccades, and they found a mix of binocular and monocular encoding, although predominantly binocular, suggesting a hybrid model between the Helmholtzian and Heringian views⁵⁰.

The second question was the focus of Dowell and colleagues⁵⁴. Following on their previous work, which discovered that larval zebrafish perform a specialized type of convergent saccade before hunting⁵⁵, the authors set out to understand how the circuits underlying saccades encode the different types of saccades. With functional imaging, they discovered that in the OMN there are two populations of motoneurons: one which is saccade-type ‘agnostic’ and another which is specifically activated during convergent, hunting related saccades. By then using laser ablations, optogenetics, and connectomics, they were able to build up a model circuit of saccadic control for both types of saccades, which involves the pretectum, the ABN and OMN, and a saccadic control region in medial rhombomeres 5 and 6⁵⁴. Interestingly, they found direct connections between the pretectal region that receives input from direction selective retinal ganglion cells (the putative homologous region to the NOT/DTN in mammals, see appendix A) and the ABN, something that was never shown in mammals, although it was reported in frogs⁵⁶.

2.3.3. Responses to optic flow: the OMR and the OKR

Zebrafish can respond to optic flow (i.e., movement of the whole visual field) with two complimentary responses: the OKR, in which the eyes track the movement of the visual field, thereby stabilizing the gaze; or the OMR, in which the fish readjusts its position with its tail, thereby stabilizing itself and its gaze. Crucially, these responses are usually selected in different contexts: when the optic flow is rotational the preferred response is the OKR, whereas when the optic flow is translational, such as when the fish is carried by a water current, the preferred response is the OMR⁶. How is translational optic flow distinguished from rotational optic flow, in order to trigger the appropriate reflex?

Since zebrafish are lateral-eyed animals, distinguishing between translational and rotational movement requires information from the two eyes to be integrated. Using functional imaging while exposing fish stimuli simulating rotational and translational flow, it was seen that in the pretectum, while there are mostly monocular responsive neurons, there is already binocular integration going on, with neurons responding specifically to translational and rotational stimuli^{10,57}.

But both the OKR and the OMR are sensory-motor responses, so it is not enough to understand the sensory end alone. What else is happening, beyond the optic flow detection, during these responses? In the case of the OMR, Naumann and colleagues⁵⁸ undertook a very complete dissection of the neural circuits responsible for the behaviour by combining behavioural experiments, functional imaging, laser ablations and circuit modelling. When it comes to the OKR, Portugues and colleagues⁸ showed a remarkably broad and distributed network of neurons underlying the response. Notably, this included neurons with strong eye position tuning in the dorsal hindbrain, a region which had not previously been implicated in the control of eye movements before. Other aforementioned studies^{50,53}, also based on optokinetic stimulation paradigms and focusing on the ventral hindbrain, revealed a high heterogeneity of neuronal responses in terms of velocity and position tuning, and monocular and binocular encoding.

A large variety of areas being activated with very heterogeneous responses make understanding the underlying processes a difficult challenge. As such, it might be more useful to abstract from the single neuron level and look at population activity⁵⁹. This was the approach followed by Feierstein and colleagues⁹, where the authors used reduced rank regression, a statistical approach that combines regression and dimensionality reduction, to analyse neural activity data from functional imaging experiments of larval zebrafish subjected to OKR and OMR evoking stimuli. Surprisingly, they found that the population activity in the hindbrain could be explained by just three modes of activity in which sensory and motor tuning was combined: one mode tuned to leftward rotation, the other to rightward rotation, and the third to translational motion⁹.

2.4. Conclusion

Simple reflexive eye movements turn out to require an elaborate control relying on sensory-motor transformations. Studying them has contributed greatly to understanding how the brain accomplishes these transformations. Larval zebrafish presents specific advantages to study visuo-motor behaviour, and indeed, from understanding the computations within the hVPNI⁴⁸, to looking at the context dependence of neuronal activity during saccades⁵⁴, to comprehensively mapping the regions underlying optic-flow responses¹⁰, there has been a great deal of progress in unravelling the neural basis of sensory-motor transformations in the oculomotor system in larval zebrafish.

Functional imaging studies have identified several regions of the larval zebrafish brain involved in sensory-motor transformations^{9,44}, and in some cases, they have allowed an in-depth comprehension of circuit function⁵⁸. Yet, there is still a lot to figure out. For example, how are these regions connected with each other? Are the neurons involved inhibitory or excitatory? When do they develop or get their functional identity? The field is rapidly progressing to answer such questions, leveraging on functional imaging and new layers of information like connectomics and genetic labelling.

In this context, the OKR appears as an experimentally tractable yet understudied model⁸. While there has been some focus in the study of saccades and of optic flow processing in the OKR, discussed above^{10,50,54,57}, the slow eye movement component of the behaviour is far less understood. Studies have shown a widespread involvement of the larval hindbrain in this reflex, including known oculomotor-related areas but also unknown neural populations with an unclear role in the behaviour^{8,9}.

The work presented in the next chapters attempts to contribute to fill in the gap regarding knowledge of the neural circuits underlying the OKR by coupling a previously established experimental paradigm⁹ to the identification of the excitatory or inhibitory nature of the discovered neural populations.

3. Methods

3.1. Line Validation

3.1.1. *In-situ* hybridization

In-situ hybridization (ISH) was based on the Hybridization Chain Reaction (HCR) 3.0 methodology⁶⁰ which features automatic suppression of background fluorescence and allows for multiplexed fluorescence ISH. The protocol was adapted from the provided molecular instruments protocol: 'HCR RNA-FISH protocol for whole-mount zebrafish embryos and larvae', with adaptations borrowed from the work of Shainer and colleagues⁶¹, and optimization tests done in the laboratory. In the day before the start of the protocol, zebrafish larvae from an *elavl3:H2B-GCaMP6s* x *vglut2a:DsRed* cross (see Appendix B for details regarding fish strains and care) at 6 days post-fertilization (dpf) were immersed in Tricaine 25X during 10 minutes in the cold and then incubated with PFA 4% in PBS (prepared with 16% Formaldehyde Solution [28909 ThermoScientific]) overnight at 4° C in a rotator. The subsequent ISH protocol spans three days and involves: 1) Dehydration with MeOH and rehydration of the fixed fish; 2) Hybridization with HCR probes for *vglut2a* and *vglut2b* transcripts and 3) Amplification with HCR hairpins labelled with the Alexa Fluor 647 fluorophore. These steps are briefly described below.

On the first day, the samples were washed with PBST (1x PBS, 0.1% Tween), then dehydrated for 10 minutes at -20°C with MeOH 100%. Afterwards, rehydration with a series of graded MeOH/PBST washes was done and the fish were pre-hybridized for 30 minutes at 37°C with probe hybridization buffer. Finally, they were incubated overnight at 37°C with the probe solution (4×10^{-9} M of *vglut2a* and *vglut2b* probes in hybridization buffer). On the second day, a series of washes with probe wash buffer and 5x SSCT (5x Sodium chloride Sodium citrate, 0.1% Tween) were done, and then, after pre-amplification for 30 minutes at 37°C with probe hybridization buffer, the fish were incubated overnight with hairpin solution (6×10^{-8} M hairpins in amplification buffer). Hairpin solution was prepared by snap-cooling the hairpins (heating 90 seconds at 95°C and cooling for 30 minutes) and adding snap-cooled hairpins to the amplification buffer. On the third and final day a series of washes with 5x SSCT were done, and the samples were then stored at 4°C.

3.1.2. Confocal image acquisition and processing

To be able to properly image the fish brain under the confocal microscope, the fishes' tail was discarded and only the head was mounted, ventral-side up, in 1.5% agarose (Agarose BP1360-100 Fisher bioreagents) in 5xSSCT on a coverslip, ensuring a minimal amount of agarose between the larva and the coverslip. A rubber ring was placed around the mounted larva with high vacuum grease and the space within it was filled with additional agarose to further stabilize the specimen. Finally, the glass slide was placed firmly on top. The prepared specimens were stored at 4°C in a humid environment.

Images were acquired using an upright Zeiss LSM 980 confocal laser scanning microscope with AiryScan 2 in Multiplex 4Y mode for acquisition at high resolution. A 40x/1.2 NA water-immersion objective was used with imaging settings to ensure maximum resolution, resulting in a voxel size of 0.065 x 0.065 x 0.25 μ m. To balance the need for very high resolution with the time-requirements of each acquisition, only a fraction of the brain was imaged, which included the

midbrain and hindbrain of one of the brain hemispheres. This part of the brain encompassed three tiles with a 10% overlap. Either the left or the right hemispheres were imaged in different samples.

The confocal image processing pipeline followed was developed and validated previously in the lab⁶² and is only briefly described in here. A more detailed description can be found in Appendix C. Essentially, it consists of three steps: pre-processing of the confocal images using FIJI⁶³, segmentation of cells based on the nuclear GCaMP signal using CellPose⁶⁴ and quantification of DsRed and ISH signals to assign cells to DsRed+ and/or ISH+ categories. Finally, by considering the ISH labelling as the ground truth of which neurons are glutamatergic, the number of True Positives (TP) (DsRed+ and ISH+), False Positives (FP) (DsRed+ and ISH-), True Negatives (TN) (DsRed- and ISH-) and False Negatives (FN) (DsRed-, ISH+) were counted and the precision, sensitivity and accuracy of the transgenic line were computed according to the following formulas:

$$(3.1) \quad \text{Precision} = \frac{TP}{TP+FP}$$

$$(3.2) \quad \text{Sensitivity} = \frac{TP}{TP+FN}$$

$$(3.3) \quad \text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

3.2. Imaging neural activity during behaviour

3.2.1. Two-photon imaging and simultaneous behaviour recording

The protocol followed in this work is largely borrowed from Feierstein and colleagues⁹. Larvae aged 6 or 7 dpf were head-restrained in agarose and imaged in a custom-built two-photon microscope.

For agarose-gel restraining of larval zebrafish, fish were placed with a Pasteur pipette in a circular 35 mm diameter petri dish with a Sylgard 184 base (Dow Corning) and the excess culture medium was removed. Afterwards, a drop of 2% low melting point agarose in E3 (Agarose BP1360-100 Fisher bioreagents) was dropped and the fish was carefully aligned in place with a carbon-fibre tweezer. After the agarose solidified, the agarose around the tail (caudal to the swimming bladder) and around the eyes was carefully removed with a fine scalpel to allow for free movements of the eyes and tail, and the fish was immersed in E3 medium. The dish with the fish was then placed onto a stage with a light-diffusing screen where the visual stimulus was projected.

The visual stimulation paradigm consisted of a pattern of radial black and red stripes projected below the fish and rotating at different angular velocities (-20°/s, -10°/s, 0°/s, 10°/s, 20°/s). The projection was divided in half so that different velocity stimuli could be projected to the left and right eyes, resulting in $5 \times 5 = 25$ different stimuli⁹. A small section in front of the stimulus was blocked to prevent unwanted stimulation of the left eye by the right hemifield stimulus or vice versa. This paradigm elicits OKR responses and leads to an enrichment in disconjugate eye movements in comparison to previously used optokinetic stimuli⁹.

Imaging frames were acquired at 2Hz and in each plane, each stimulus was presented, in a pseudo-random order, for 5 seconds, resulting in 125s of imaging time per plane. 180 to 200 planes, spaced 1 μm apart, were imaged to achieve a pseudo-volumetric imaging, i.e., the whole volume is imaged but not simultaneously. The resulting imaged volume had a voxel size of $0.57 \times 0.57 \times 1 \mu\text{m}$ and spanned $411 \times 265 \times 180 \mu\text{m}$, capturing the whole hindbrain as well as parts of the caudal

midbrain. The imaging region was chosen so as to include most oculomotor related areas, as well as the whole ventral-dorsal extension of the hindbrain.

Fluorescence data was captured using two photomultipliers (PMT): one for capturing green fluorescence from the GCaMP signal and the other for capturing red fluorescence from the DsRed signal. The functional experiment was done with the green PMT, but at the end of each functional imaging experiment an anatomical stack of the same area as the functional stack was acquired with both the green and red PMTs, to retrieve information on the DsRed labelling of cells.

3.2.2. Behaviour data processing

Eye and tail position data were extracted online and processing of the behaviour-tracking data used custom-made MatLab scripts, as described elsewhere⁹. The eye data processing went through slight alterations, so it is described below.

Eye position traces were smoothed using a Savitzky-Golay polynomial filter (order 5, 150ms width) and a median filter (50ms width). Eye velocity was estimated from the unfiltered position trace using a differentiator filter (order 10, passband frequency 0.15Hz, stop band frequency 0.2Hz) and it was filtered with a Butterworth filter (order 5, normalized cutoff frequency 0.0667). Saccades were extracted from the velocity trace by choosing points where the velocity was above three standard deviations of the average velocity and setting the remainder of the trace to zero. A slow velocity trace was constructed by first removing the saccades and replacing the values using the ‘fillmissing’ (‘nearest’) function in MATLAB and then applying a Butterworth filter again (order 5, normalized cutoff frequency 0.0667).

3.2.3. Functional image processing

To process the image data from functional experiments, a Python-based pipeline was developed. The schematic diagram of the pipeline is depicted in Figure 3.1. Briefly, frames from each plane were motion corrected using suite2p’s motion correction algorithm⁶⁵ and then planes were aligned to each other. Afterwards, to identify putative neurons, the average intensity projection image of each plane was used to segment the neuron’s nuclei with StarDist, a deep learning tool for nuclei segmentation⁶⁶, with a custom trained model. The calcium trace of each 2D ROI (Region of Interest) was computed as the average fluorescence of the pixels within the ROI over time. 3D neurons were constructed from the 2D segmented data by stitching overlapping neurons across planes. The final calcium trace was calculated by calculating the DF/F of each constituent 2D ROI, concatenating and z-scoring the result. These steps are described in greater detail in Appendix D.

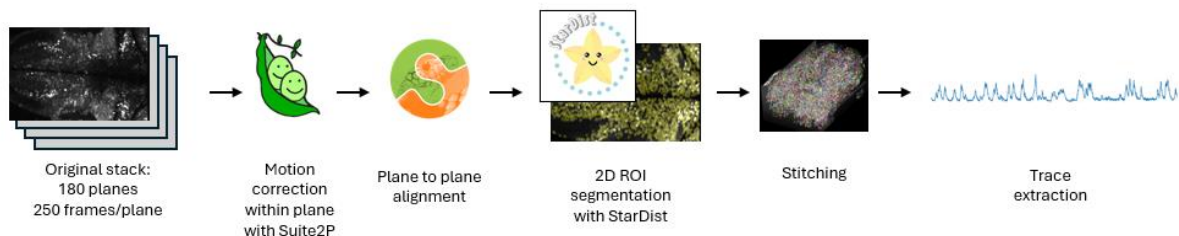


Figure 3.1. Schematic of the python pipeline developed in this work to process functional imaging datasets.

3.3. Data analysis of functional imaging experiments

3.3.1. Identification of neuronal neurotransmitter identity

Under the experimental design followed in the experiments presented here, identifying cells as glutamatergic or GABAergic depends on first identifying cells in the functional dataset, and then checking, for each of these identified cells, its DsRed labelling by looking into the red channel of the anatomical stack taken after each experiment. However, since the experiments are quite long, there is always some distortion of the fish, or changes in its position, that make the anatomical and functional data misaligned. To overcome this, the anatomical stack, after being motion-corrected with custom-made MatLab scripts, was registered to the functional stack using the Advanced Normalization Tools (ANTs) package⁶⁷.

After the images were aligned, the masks computed in the functional data were applied and a similar approach as the one used for the confocal image processing was used to classify cells as DsRed positive, namely, the number of very bright pixels (above 90% percentile of intensity values per slice) was calculated for each neuron, and a threshold was manually chosen to classify cells as DsRed+ or DsRed- (between 105 and 115 for the different fish).

3.3.2. Building an anatomical reference brain

While inspecting data from each fish is highly informative, it is useful to leverage this data and look at the pooled neural data from multiple fish from the same or different experiments⁸. To do so, the different fish brains must be aligned, using image registration algorithms, to a common reference. For this purpose, a reference brain was built. The methodology to do so is described in detail in Appendix E.

3.3.3. Correlation Analysis

To understand how neural activity relates to oculomotor behaviour, a correlation analysis was performed^{8,12,53}. A set of behavioural and stimulus related regressors was built from the processed behaviour data. Eye position and eye velocity data were median centred and rectified. Non-rectified eye position and eye velocity data were also kept as regressors – see regressor list in Table 3.1. All regressors were then convolved with a GCaMP kernel with the formula:

$$(3.4) \quad G6s(t) = A(e^{\frac{-t}{\tau_{off}}} - e^{\frac{-t}{\tau_{on}}})$$

Where $\tau_{off} = 2.392s$, $\tau_{on} = 2.179s$, and A is a normalizing constant so that the integral of the function is one (time constants estimated from *in vivo* measurements by Aaron Ostrovsky). This convolution emulates the slow rise and decay of the GCaMP6s signal, therefore giving an estimate of the predicted neuronal calcium activity response to each behaviour/stimulus⁴⁴. An additional ‘motion-artefact’ regressor was created⁵⁴, by creating a binary vector which was set to one during the frames with too much motion, as identified during the motion correction.

Each neuron’s trace was then correlated with each regressor in Python. To understand if the obtained correlation values were significant, an empirical null distribution of maximum correlation values per neuron was created by circularly shifting the regressors 100 times by a random number of

frames and recalculating the correlations. A Kolmogorov Smirnov test was used to attest the difference between the empirical null distribution and the observed distribution of maximum correlations. The 95th percentile of this empirical null distribution (around 0.25 for the different fish) was used to establish a first threshold on what neurons to keep for further analysis. Neurons whose maximum correlated regressor was the motion artefact regressor were also removed from the analysis. A higher threshold of 0.4 was then selected heuristically to plot individual correlation maps. The choice of threshold is not critical in the sense that the populations of neurons that are seen in the brain as related to different variables are robust to small variations in this threshold. Correlation maps were plotted in 2D using the package matplotlib⁶⁸. In every map, an anatomical compass indicates the Posterior (P), Anterior (A), Left (L), Right (R), Ventral (V) or Dorsal (D) directions.

3.3.4. Eye position and eye velocity tuning

Distinguishing between eye-position coding and eye-velocity coding neurons is an important but challenging task due to the high correlation between these variables. To do so, a set of regression models were fitted to each neuron to calculate the Explained Variance (EV) of eye position and eye velocity. For this analysis, non-rectified regressors related to eye position, eye velocity and stimulus were used (LEP, LEV, LES, REP, REV RES, see table 3.1). For example, to calculate the explained variance of left eye position, each neuron was fit with the following pair of models:

$$(3.5) \quad \text{Full Model (FM): } activity = \beta_1 LE_{position} + \beta_2 LE_{velocity} + \beta_3 LE_{stimulus} + \varepsilon$$

$$(3.6) \quad \text{Reduced Model (RM): } activity = \beta_1 LE_{velocity} + \beta_2 LE_{stimulus} + \varepsilon$$

The explained variance was then computed as:

$$(3.7) \quad EV(LE_{position}) = \Delta R^2 = R_{FM}^2 - R_{RM}^2$$

The ΔR^2 is used as a proxy for explained variance of each variable because it reflects the explanatory contribution unique to that variable, in the sense that it is the contribution of the part of the variable which is orthogonal, or decorrelated, with the other variable^{69,70}. If a neuron responds solely to left eye velocity, then $LE_{position}$, even if it is correlated with activity as a side-effect of the position-velocity correlation, will not add any explanatory power to the model: ΔR^2 will be low. On the other hand, if the neuron really responds to left eye position, the full model will explain neural activity much better than the reduced model, and therefore ΔR^2 will be high.

The same was done for left eye velocity, right eye position and right eye velocity. Then, for each neuron a unique pair of $EV(E_{position})$ and $EV(E_{velocity})$ values were attributed by choosing the preferred eye of that neuron – i.e., the side to which neural activity had maximum correlations with. Finally, a Position-Velocity index (PV index) was calculated with the following formula:

$$(3.8) \quad PV \text{ index} = \frac{EV(E_{position}) - EV(E_{velocity})}{EV(E_{position}) + EV(E_{velocity})}$$

This formula was modified from the work of Brysch and colleagues⁵³, in which the authors calculated a similar PV index, but with correlation values instead. This modification was necessary because, in that work, the authors used a specific protocol to decorrelate eye position and eye velocity, and therefore did not have the inherent mixture between position and velocity that exists in the data presented here. In addition, the stimulus regressors were included in both the reduced and full models in an attempt to tackle the stimulus-velocity correlation.

This approach relies on two important assumptions. The first one is the assumption of monocularity. In every step of the analysis, responses are considered as related to left or right eye

variables. However, it is known that neurons often respond binocularly^{9,53}, so it might be misleading to infer their tuning from looking at single eye responses alone. The second assumption is that the decorrelated components of correlated variables are neurobiologically relevant. In highly correlated variables, such as eye velocity and stimulus velocity, the component of one orthogonal to the other might not have explanatory power not because the neuron is not responding to both variables, but because this component might not represent any behaviourally interpretable variable. In that case, the approach used here does not necessarily equate to separating their contributions in the neural responses - sometimes this may be impossible to achieve.

Table 3.1. Regressors used in the correlation analysis.

Regressor	Short name
Left Eye Position	LEP
Right Eye Position	REP
Left Eye Position Temporal Rectified	LETP
Left Eye Position Nasal Rectified	LENP
Right Eye Position Temporal Rectified	RETP
Right Eye Position Nasal Rectified	RENP
Left Eye Velocity	LEV
Right Eye Velocity	REV
Left Eye Velocity Temporal Rectified	LETV
Left Eye Velocity Nasal Rectified	LENV
Right Eye Velocity Temporal Rectified	RETV
Right Eye Velocity Nasal Rectified	RENV
Left Eye Stimulus Velocity	LES
Right Eye Stimulus Velocity	RES
Left Eye Stimulus Velocity Clockwise Rectified	LENS
Left Eye Stimulus Velocity Counterclockwise Rectified	LETS
Right Eye Stimulus Velocity Clockwise Rectified	RETS
Right Eye Stimulus Velocity Counterclockwise Rectified	RENS
Stimulus Clockwise Rotation	Stim clock
Stimulus Counterclockwise Rotation	Stim counterclock
Stimulus Convergence	Stim Conv
Stimulus Divergence	Stim Div
Motion Artifact	motion artifact

4. Results

4.1. Fidelity of the transgenic vglut2a:DsRed line

To assess the fidelity of the vglut2a:DsRed line in reporting glutamatergic neurons, an *In-Situ* Hybridization (ISH) with probes for the vglut2a and vglut2b genes was done, followed by high-resolution confocal imaging. The ISH labelling serves as a proxy for the true glutamatergic neurons. Four vglut2a:DsRed x elavl3:H2B-GCaMP6s fish were imaged, and the imaged region spanned the hindbrain and midbrain of one brain hemisphere, with an average depth of 134.13 μm (Figure 4.1A).

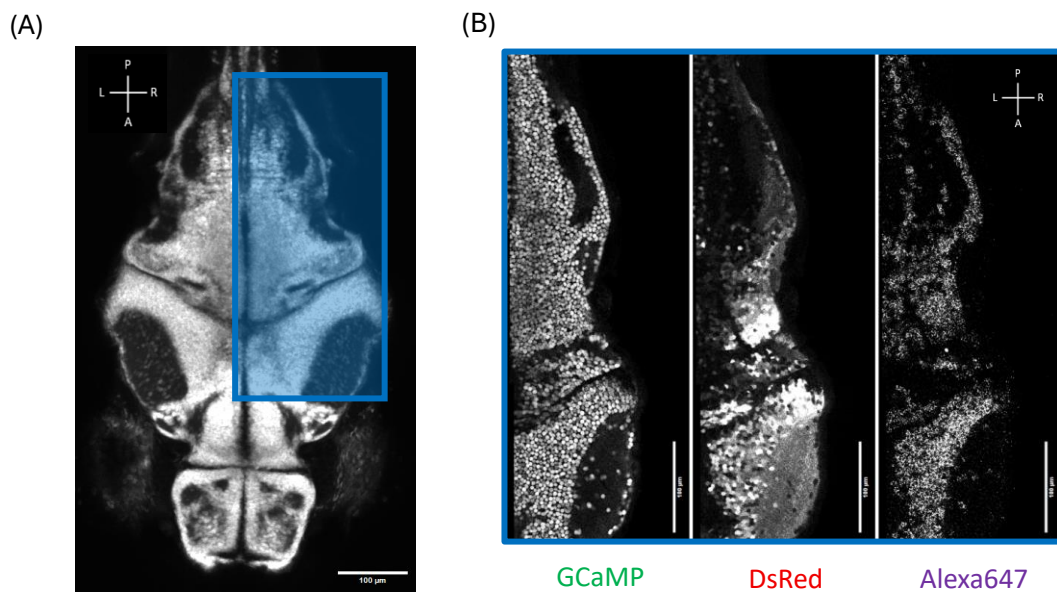


Figure 4.1. A representative example of a confocal imaging experiment. (A) Imaged region (blue rectangle) represented on a plane from the Z-brain atlas⁷¹. (B) From left to right, the three markers are shown for an example slice: nuclear GCaMP (green), DsRed expression under the vglut2a promoter (red) and ISH labelling of glutamatergic neurons (far-red).

4.1.1 Qualitative overview of line fidelity

By comparing the ISH labelling and the DsRed expression, one can find cells labelled in both channels (glutamatergic cells), in none (non-glutamatergic cells), just labelled by the ISH (glutamatergic cells missed by the line) or just labelled by the DsRed expression (non-glutamatergic cells labelled by the line) (Figure 4.2). It is worth noting that the ISH labelling is spatially different from the DsRed labelling: while the latter homogeneously fills the neuron somas, the former appears as bright dots scattered in the reduced and peripheral cytoplasm of the neurons.

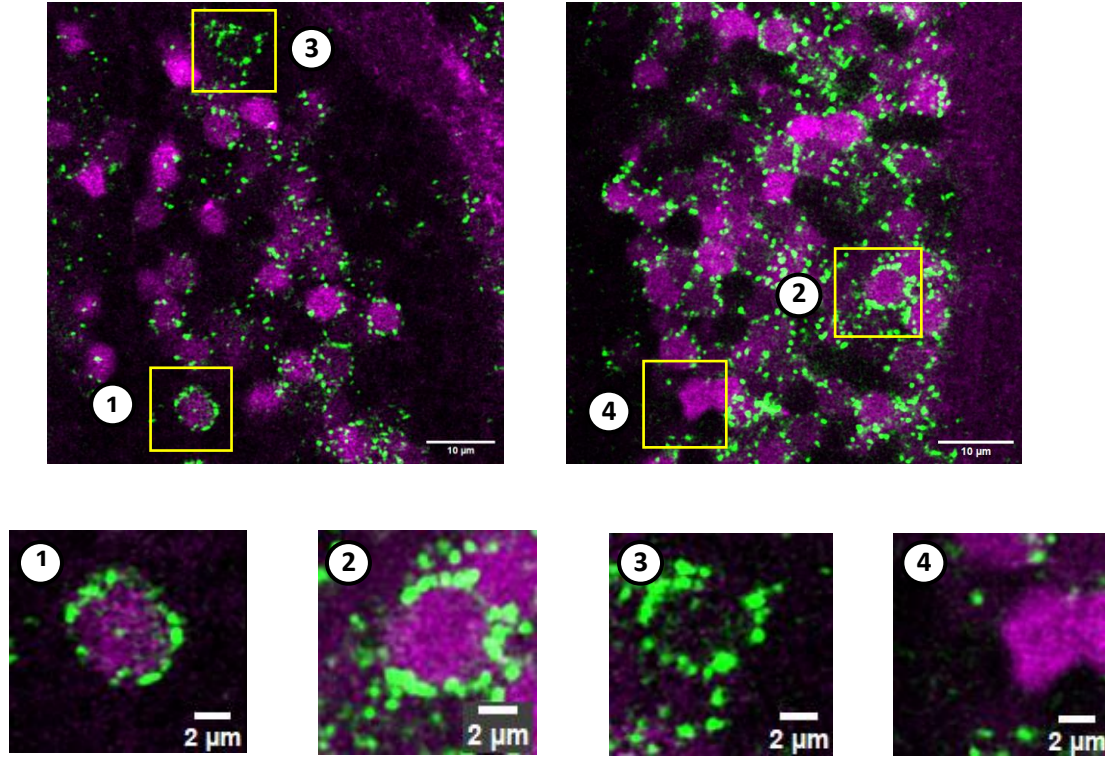


Figure 4.2. ISH versus DsRed labelling. (Top) Regions from an example slice of the multichannel stack for one fish. Green corresponds to the ISH labelling and magenta to the DsRed expression under the *vglut2a* promotor. Yellow squares highlight cells displayed below. (Bottom) Examples of four neurons, including two ISH+ DsRed+ (1 and 2), one ISH+ DsRed- (3) and one ISH- DsRed+ (4 - in the lower right).

Overall, a qualitative comparison seems to show a high agreement between the two channels, as can be seen in the example slice shown in Figure 4.1B.

4.1.2. Quantitative characterization of line fidelity

To move from a subjective analysis to a more objective and quantitative assessment, an automatic segmentation and classification pipeline was applied to the images. The results for each fish are listed in Table 4.1. On average, the transgenic line had a $74 \pm 3.05\%$ precision (i.e., 74% of cells labelled as DsRed+ are ISH+), a $62 \pm 2.06\%$ sensitivity (i.e., out of the ISH+ cells, 62% are DsRed+) and a $71 \pm 2.63\%$ accuracy (see formulas in section 3.1.2). An example of the classification can be seen on appendix F – Figure F.1. While it seems to be an accurate classification overall, weakly-labelled DsRed cells are often taken as DsRed-negative, and apparently ISH-negative cells adjacent to ISH-positive ones are sometimes taken as ISH-positive, adding some uncertainty to the results.

Table 4.1. Results from the quantitative validation of the transgenic line. For each imaged fish, the computed Precision, Sensitivity and Accuracy of the transgenic line is shown, as well as the imaged hemisphere. On the bottom lines the average and standard deviation (STD) of each metric for the four fish is shown.

ID	Hemisphere	Precision	Sensitivity	Accuracy
<i>Fish 1</i>	Right	0.7364	0.6438	0.7277
<i>Fish 2</i>	Left	0.7567	0.6283	0.7119
<i>Fish 3</i>	Right	0.7761	0.6265	0.7167
<i>Fish 4</i>	Left	0.7048	0.5947	0.6678
<i>Average</i>	-	0.74	0.62	0.71
<i>STD</i>	-	0.03	0.02	0.03

4.2. Neural correlates of oculomotor behaviour

A series of functional imaging experiments were done with double transgenic fish (elavl3:H2B-GCaMP6s x vglut2a:DsRed / x gad1b:DsRed) to probe into the neural correlates of oculomotor behaviour. Out of the seven vglut2a and seven gad1b fish datasets that were obtained, four vglut2a and three gad1b fish were chosen. This choice was motivated by the quality of the datasets, namely, datasets with motion distortion, aberrant tracking data, or, in the case of the gad1b fish dataset, few neurons correlated with the regressors, were excluded. The whole dorsoventral extension of the hindbrain and caudal midbrain were imaged in the green channel while exposing the fish to optokinetic stimuli. The imaged region can be seen in Figure 4.3A. Afterwards, an anatomical stack of the GCaMP and DsRed channels was acquired (Figure 4.3B). The neurons were segmented and activity traces from each segmented neuron were extracted (Figure 4.3C). Gad1b fish had a lower signal intensity and fewer segmented neurons than vglut2a fish (appendix F, Figure F.2). The activity traces were then correlated with several stimuli and behaviour derived regressors (Table 3.1).

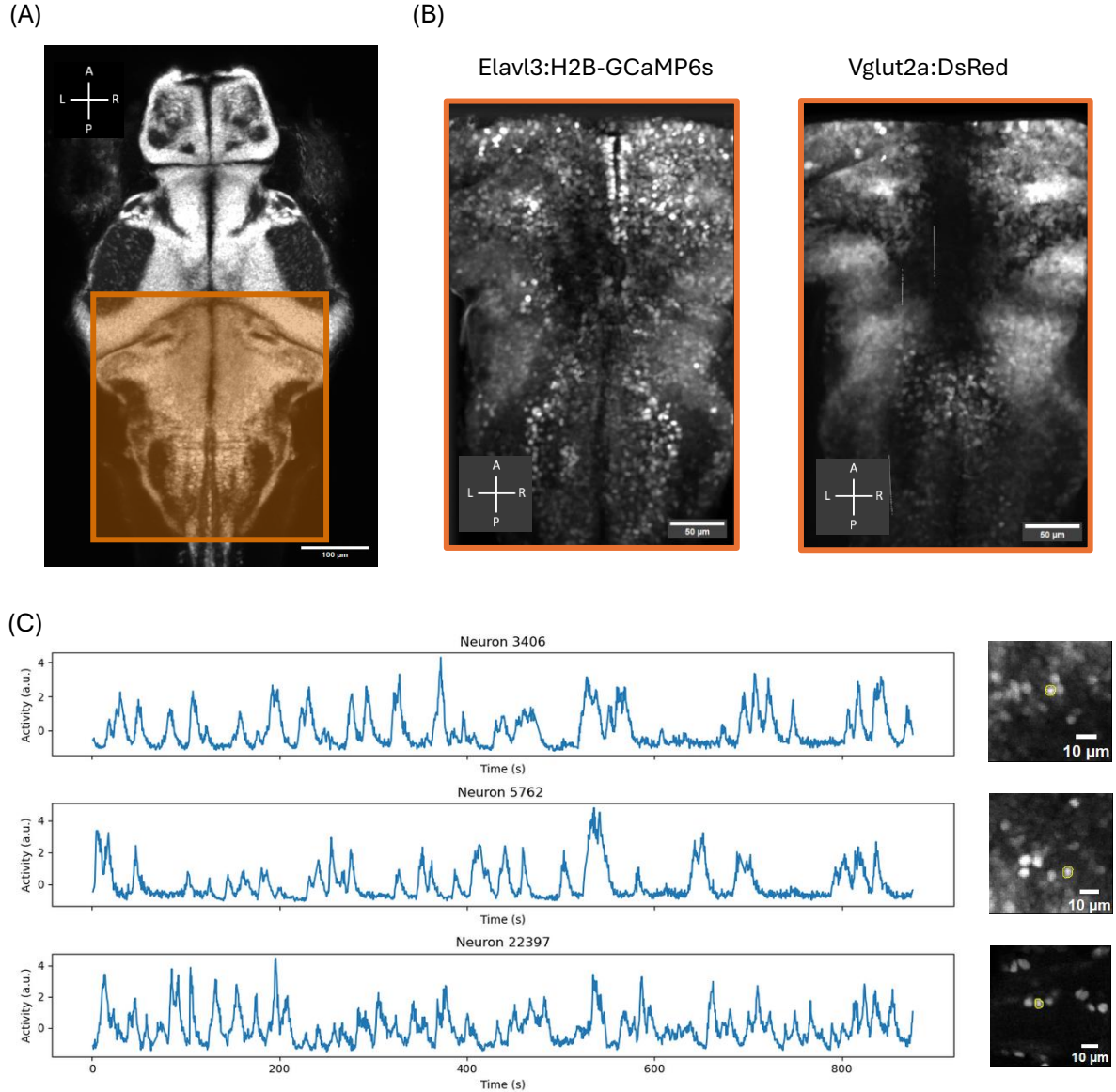


Figure 4.3. A representative example of a functional imaging experiment. (A) Imaged region (orange rectangle) represented on a plane from the Z-brain atlas⁷¹. (B) Standard-deviation projections of the green channel (calcium activity signal) (left) and red channel (DsRed labelling of glutamatergic neurons) (right). (C) Example traces (left) from segmented neurons shown on the right.

4.2.1. Correlations between neural activity and oculomotor variables

To understand the significance of the observed correlations, an empirical null distribution of maximum correlation values was calculated. The distribution of maximum correlation values per neuron (i.e., the maximum correlation value a neuron has with any one of the behavioural variables) shows a pronounced rightward skew when compared with this null distribution (Figure 4.4A, data for all fish can be found in appendix F - Figure F.3). For every fish, the empirical null distribution was significantly different from the real data distribution at $p < 0.001$ (Kolmogorov-Smirnov test), indicating that the observed correlations are not random artefacts.

Indeed, a maximum correlation value above 0.4 is observed 61 more times in the real data than in the control data. Moreover, by looking at some neural activity traces and their corresponding

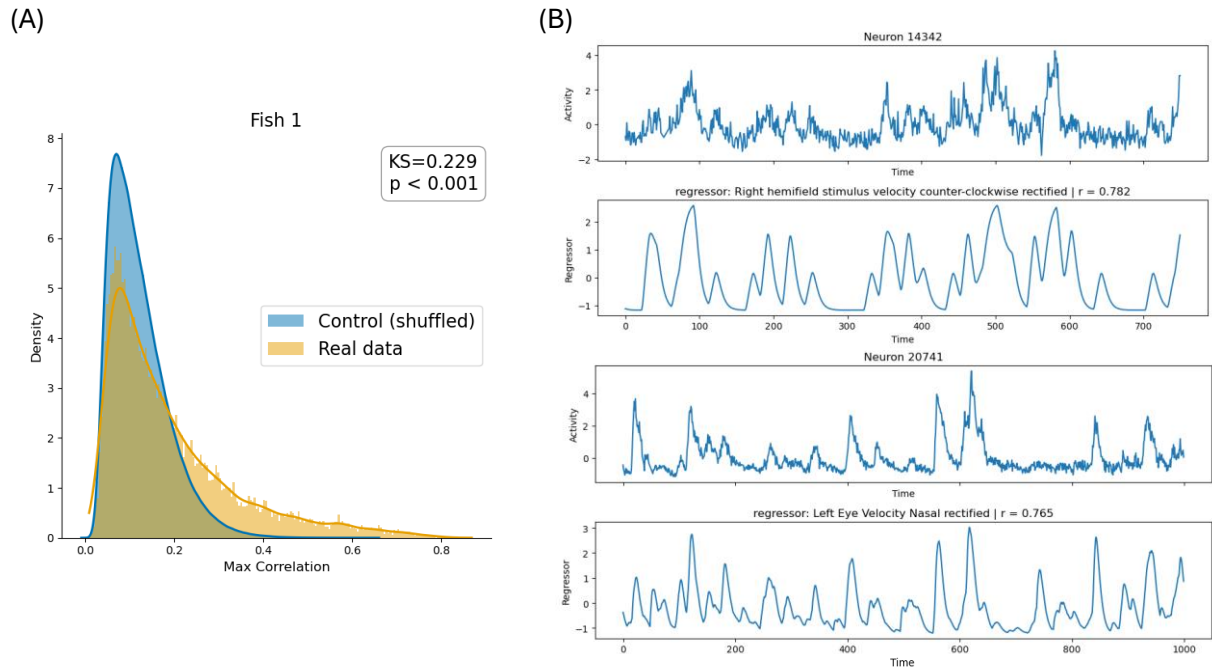


Figure 4.4. Observed correlations are significant. (A) Density histograms of the maximum correlation value per neuron for control (blue) and real (yellow) data. The lines are kernel-density estimates of the continuous density. On the top right corner, the Kolmogorov-Smirnov test statistic and p-value are shown. (B) Examples of two neurons' traces along with their maximum correlated regressor. Notice how similar the neural activity is to the behavioural or stimulus related regressors.

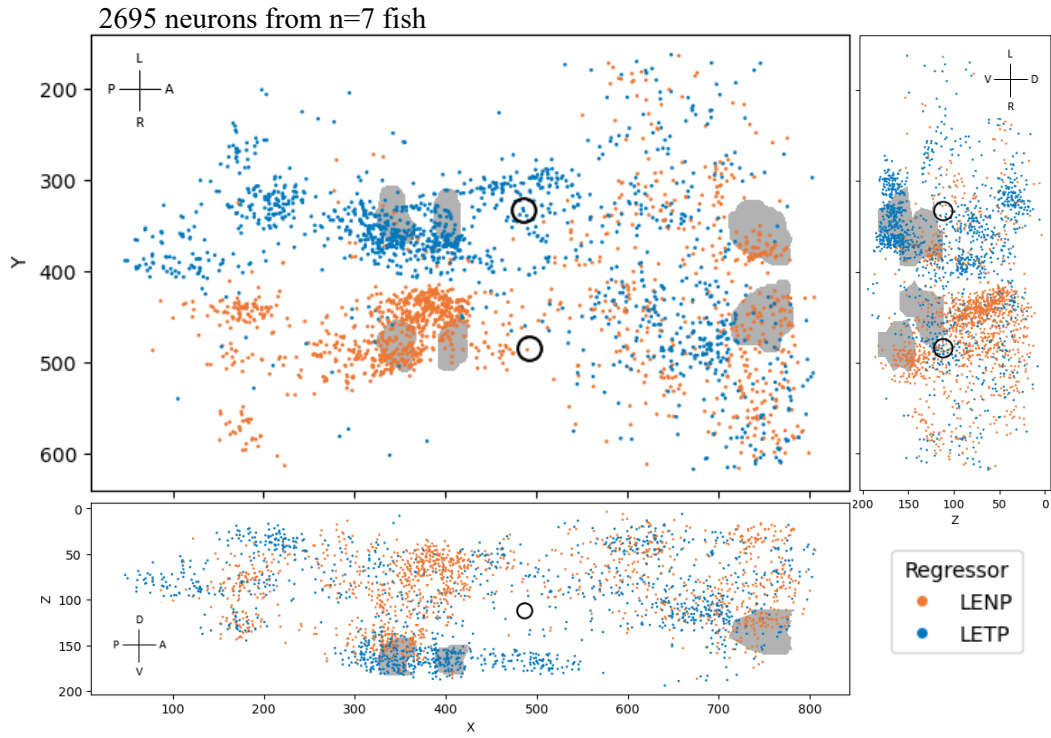
top-correlated regressor, one can see that there are neurons whose activity closely follows behavioural or stimuli variables (Figure 4.4B).

4.2.2. Spatial distribution of oculomotor-tuned neurons

Having concluded that some of the segmented neurons are in fact correlated with the behaviour and the stimulus, a question immediately arises: how are these correlations structured spatially? To probe into the spatial organization of the neurons, a series of correlation maps were produced where the neuron centroids were coloured according to their maximum correlated regressor. The analysis that ensued was restricted to two important oculomotor variables: eye position and slow eye velocity (i.e., eye velocity excluding the saccadic component). In Figure 4.5 and 4.6 the correlation maps for eye position and eye velocity, respectively, are shown.

Both maps reveal a widespread and highly symmetrical distribution of oculomotor-tuned neurons in the hindbrain and caudal midbrain. (Figures 4.5 and 4.6). Neurons seem to be overwhelmingly tuned to ipsiversive eye movement and position, i.e., in the left hemisphere they respond to leftward movement and in the right hemisphere to rightward movement.

Left eye position



Right eye position

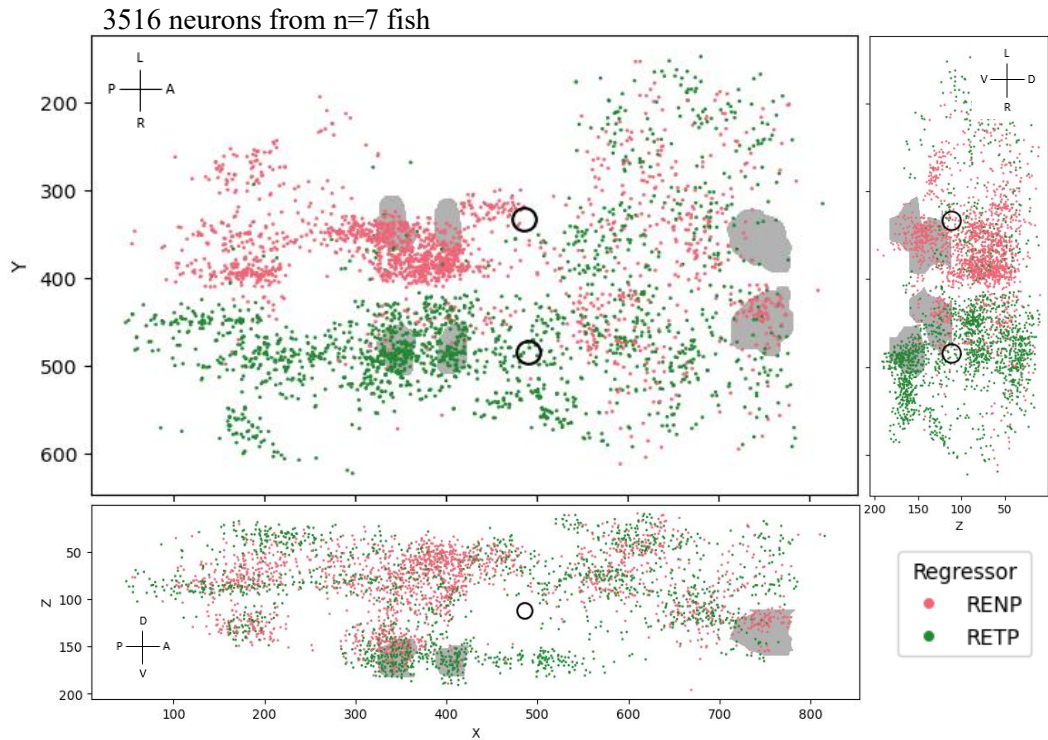


Figure 4.5. Eye position correlation maps. Data from n=7 fish. Each point is a neuron centroid; its colour corresponds to a regressor according to the legend (see abbreviations in Table 3.1). (Top) Left eye position map. (Bottom) Right eye position map. In each map, a horizontal (top), coronal (right) and sagittal (bottom) view of the brain is shown. Black circles indicate the location of the Mauthner cell. Shaded grey regions correspond to the ABN and OMN (see appendix E).

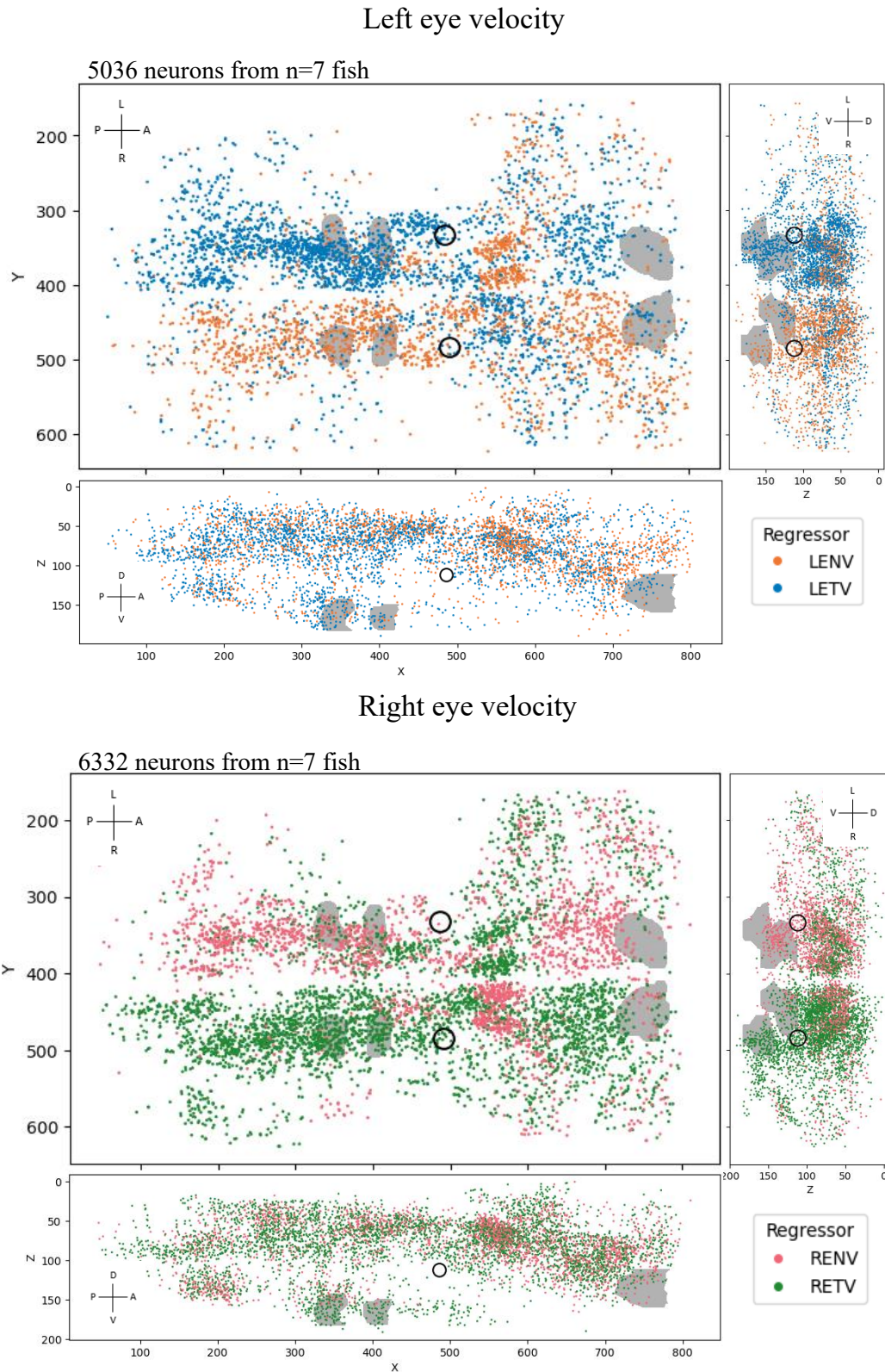


Figure 4.6. Slow eye velocity correlation maps. Data from n=7 fish. This figure is similar to the previous one. Each point is a neuron centroid; its colour corresponds to a regressor according to the legend (see abbreviations in Table 3.1). (Top) Left eye velocity map. (Bottom) Right eye velocity map.

On the eye position map, neurons separate clearly not only according to leftward versus rightward position but also according to nasal versus temporal position. Namely, in the region of the abducens nucleus (ABN), temporal tuned neurons are more ventral than nasal tuned ones (Figure 4.5A, sagittal view. See also Appendix F – Figure F.4 for a coronal view with the four types of neurons). In the oculomotor nucleus (OMN), although there are few neurons overall (because this region was often missed in the imaging), neurons seem to be predominantly nasal tuned. In the middle part of the dorsal hindbrain there are many nasal tuned neurons but in the more caudal, lateral area there is a small cluster of temporal responsive cells (Figure 4.5, coronal and horizontal views). On the other hand, the optic tectum and cerebellum show a salt-and-pepper distribution with no discernible structure.

Looking at the eye velocity map, the first noticeable difference is that there appear to be more eye velocity tuned neurons than eye position tuned neurons. The dorsal hindbrain is specially enriched in eye velocity tuned neurons. The nasal versus temporal distinction is also less clear, with nasal and temporal tuned neurons intermingled in every area. A notable exception is the ABN region, where neurons are predominantly temporal tuned. Overall neurons seem to be mostly structured according to leftward versus rightward movement. Yet, despite the general ipsiversive tuning, there is a compact pair of medial clusters of cells organized into two stripes, probably corresponding to the hindbrain oscillator (HBO)⁷², that appears to be tuned to contraversive eye movements.

4.2.3. Eye position and velocity encoding regions

To further explore the differences between the eye position and eye velocity tuning observed in the correlation maps, a position-velocity index was computed (see methods, section 3.3.4). The spatial distribution of neurons according to their PVI value can be seen in the PVI map in Figure 4.7.

The PVI map reveals that there are distinct regions in the hindbrain tuned to eye position or eye velocity. Namely, in the ventral hindbrain it is possible to view a large cluster of position tuned neurons in and around the ABN, as well as a caudal cluster of velocity tuned neurons in the IO and a rostral, lateral cluster of position tuned neurons. In the dorsolateral caudal hindbrain there is a small cluster of position neurons that correspond to the temporal tuned neurons described in the above section. The HBO also appears saliently as a velocity tuned cluster. In the remaining regions, including the cerebellum, optic tectum and middle of the hindbrain, the distribution is less structured and more salt-and-pepper like.

This observation led to the manual outlining of seven regions based on PVI values (Figure 4.8).

Four predominantly eye-position coding clusters were found: The region including and surrounding the ABN masks ('ABNregion', average PVI of oculomotor tuned neurons in this cluster = 0.42), the OMN ('OMN', average PVI = 0.24), a narrow strip in the ventral lateral middle hindbrain region (here named middle hindbrain position nucleus - 'MHbpos', average PVI = 0.70), and the dorsolateral caudal clusters mentioned above (here named caudal dorsal position nucleus - 'CDpos', average PVI = 0.24).

Two strongly velocity preferring nuclei were also found: the inferior olive ('IO', average PVI = -0.71) and the HBO ('HBO', average PVI = -0.77).

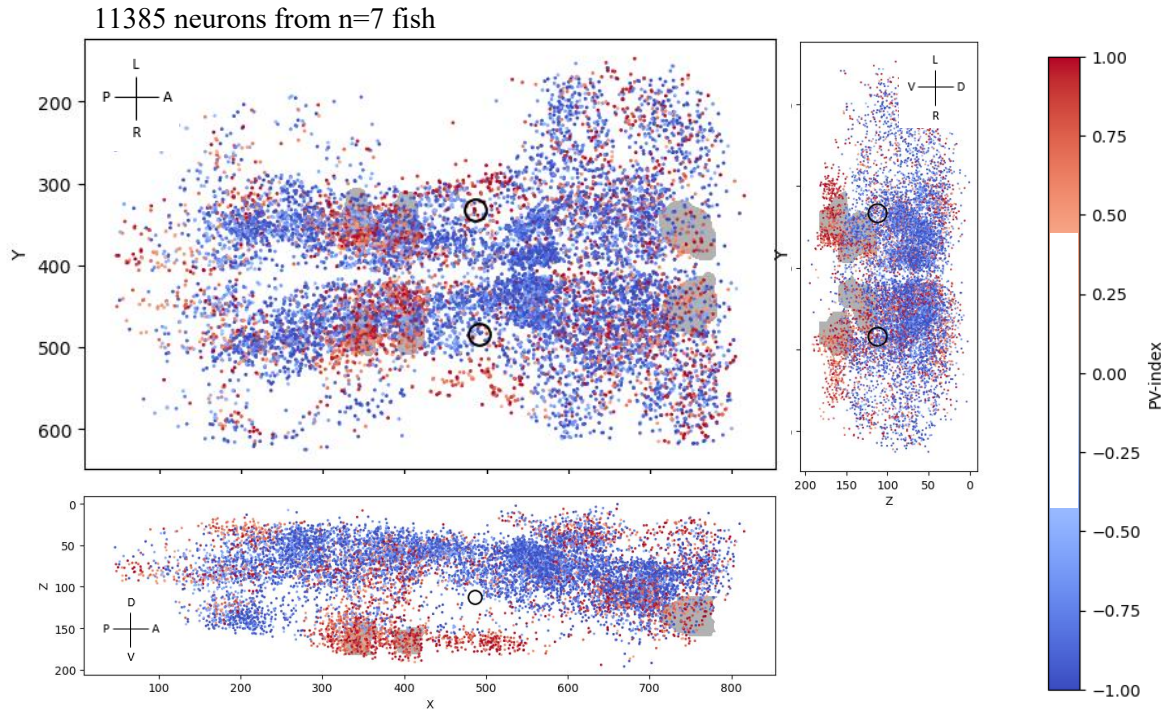
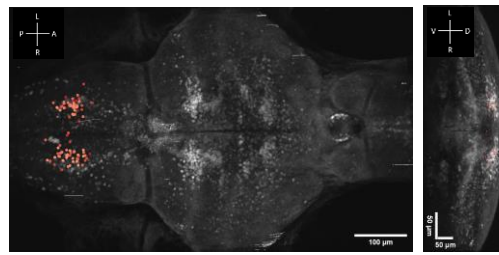


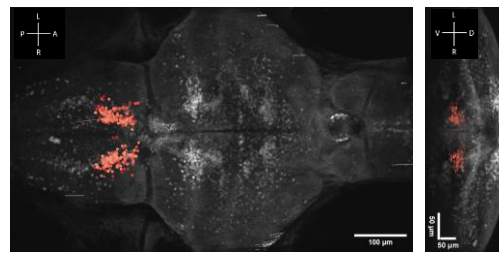
Figure 4.7. Position-velocity index map. This figure is similar to Figure 4.6, but for each neuron (from n=7 fish), its colour corresponds to its PVI according to the colour bar shown on the right. Intermediate values ($-0.4 \leq PVI \leq 0.4$) are masked. Blue corresponds to strongly velocity preferring neurons while red corresponds to strongly position preferring neurons.

Finally, a ‘mixed’ nucleus was also found: a very compact group of cells in the ventral caudal hindbrain, lateral and slightly caudal in relation to the IO (here named caudal lateral mixed nucleus - ‘CLmix’, average PVI = -0.03). This nucleus was easily distinguishable and included both position tuned cells, velocity tuned cells, and position-velocity tuned cells. Note that this region is not evident in Figure 4.7 because neurons with intermediate PVI are masked.

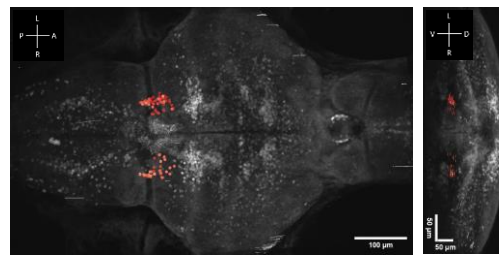
Eye position clusters



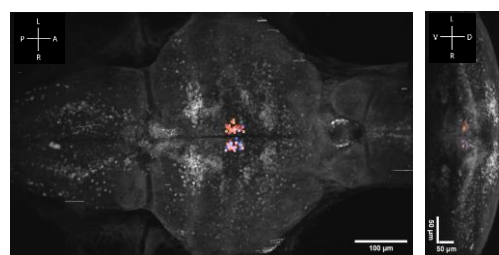
Caudal Dorsal Position nucleus (Cdpos)



Abducens nucleus region

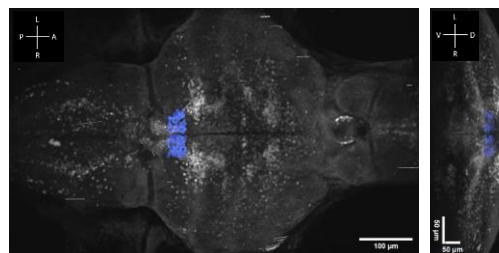


Middle Hindbrain Position nucleus (MHbPos)

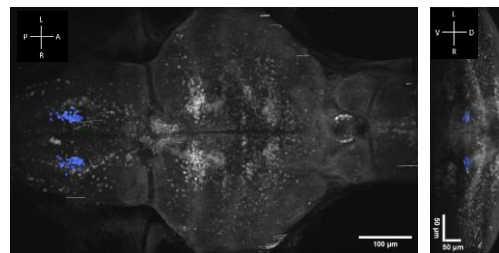


Oculomotor nucleus

Eye velocity clusters

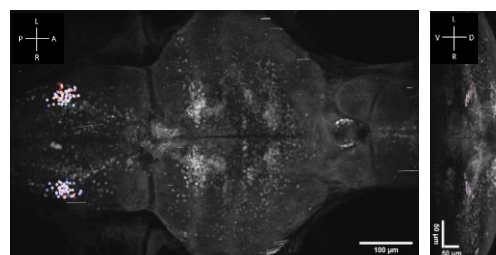


Hindbrain oscillator



Inferior olive

Mixed cluster



Caudal Lateral mixed nucleus (CLmix)

Figure 4.8. Manually outlined neuron clusters. Each image shows a horizontal and coronal standard-deviation projection image of the reference brain (see appendix D) with the neuron centroids from the attributed cluster from $n=7$ fish overlain, coloured according to their PVI.

4.2.4. Neurotransmitter identity of outlined eye position and velocity clusters

After outlining the eye position and eye velocity clusters as described above, the neurotransmitter identity of the neurons within each cluster was determined by looking at the percentage of *gad1b*⁺ or *vglut2a*⁺ neurons in each cluster, for each fish. The automatic classification of cells as DsRed⁺ or DsRed⁻ was validated by manually annotating three slices and comparing the annotation to the classification (appendix F – figure F.5). The classification method has a precision of 84% (i.e., 84% of cells labelled as DsRed⁺ are DsRed⁺) and a sensitivity of 62% (i.e., out of the DsRed⁺ cells, 62% are labelled as such).

The results for the cluster identities are presented in Figure 4.9. The IO was the only cluster whose neurotransmitter identity could be resolutely determined, with around $57.9 \pm 14.5\%$ of cells being putatively glutamatergic. In the remaining clusters, the DsRed labelling was less present. Still, the CDpos and the HBO had relevant GABAergic components ($35.5 \pm 10\%$ and $17.5 \pm 8.5\%$, respectively) and the ABNregion and the CVMix had relevant glutamatergic components ($24.4 \pm 2.7\%$ and $9.1 \pm 7.8\%$, respectively). The distribution of GABAergic and glutamatergic neurons within the clusters appears to be largely disjoint, with no cluster including appreciable portions of both types of cells.

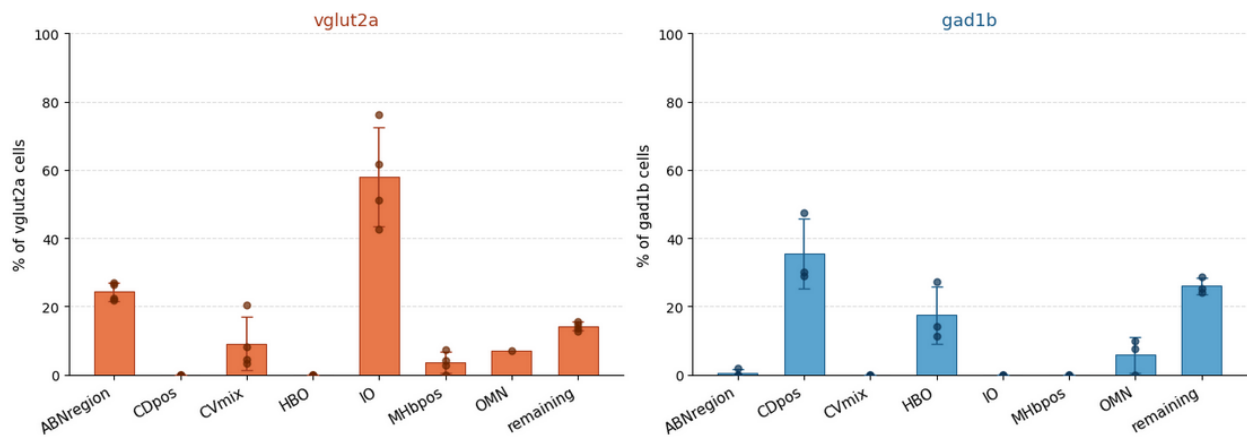


Figure 4.9. Neurotransmitter composition of the neural clusters. Percentage of putatively glutamatergic (left, orange) or GABAergic (right, blue) neurons within each of the manually outlined clusters. Bars show the average plus or minus standard deviation of $n=4$ fish for the *vglut2a* and $n=3$ fish for *gad1b*.

5. Discussion

5.1. Overview

In this thesis, imaging of brain activity of larval zebrafish exposed to optokinetic stimulation was coupled with the identification of the neurotransmitter identity of the imaged neurons, with the goal of understanding the neural correlates of the sensory-motor transformations underlying oculomotor behaviour.

A series of functional imaging experiments were done with fish expressing a genetically encoded calcium indicator pan-neuronally and a fluorophore under the control of either the *vglut2a* or *gad1b* regulatory region, presumably targeting glutamatergic and GABAergic populations, respectively. The fidelity of the *vglut2a* line, i.e., how well it reports the glutamatergic phenotype, was assessed by quantitatively comparing the line expression pattern with an ISH staining, and an estimated precision of 74% and sensitivity of 62% were obtained.

The imaging datasets were processed with a custom-made python pipeline to extract neuronal activity data, which was then correlated with a suite of oculomotor and stimuli related variables. A widespread and highly structured distribution of eye position and eye velocity tuned neurons was found. The properties of these neurons were further explored by computing a position-velocity index (PVI), which separated neurons into clusters of neurons tuned to eye position or eye velocity.

Seven neuronal populations of interest were then manually outlined and the percentage of putatively glutamatergic or GABAergic cells within each one was computed. Some of these populations seem to correspond to well-known oculomotor structures (such as the ABN, OMN, IO and HBO) but there were also three unknown or less characterized groups (here designated MhbPos, CDpos and CVmix). In this section, the results of the project will be discussed.

5.2. Validating the transgenic lines used in this work

Two transgenic lines were used for determining neurotransmitter identity: *vglut2a*:DsRed and *gad1b*:DsRed, which label putative glutamatergic and GABAergic neurons, respectively. Both lines were created by Bacterial Artificial Chromosome (BAC) mediated transgenesis, and the authors did not undertake a quantitative assessment of their fidelity^{73,74}. Nonetheless, they have been widely used, and are often taken to be direct proxies for neurotransmitter identity^{39,75,76}.

The quantitative validation presented in this work revealed the *vglut2a* line to have a precision of 74% and a sensitivity of 62% in reporting glutamatergic identity, while previous work in the lab has shown for the *gad1b* line a precision of 81% and sensitivity of 56%⁶².

What could explain the obtained estimates? The low sensitivity was expected since the ISH hybridization is done with *vglut2a* and *vglut2b* probes. *Vglut2b* is a marker for a distinct set of glutamatergic cells in zebrafish and as such the *vglut2a* line is not expected to label *vglut2b* glutamatergic neurons. A parallel situation occurs for the *gad1b* line, where *gad1b* and *gad2* probes were used. The relatively low precision is harder to explain, but it is probably because, in BAC-mediated transgenesis, the construct is randomly inserted in the genome, with variable copy-

number⁷⁷. As such, in comparison to the actual *vglut2a* gene, it will be under the influence of a different chromatin structure, moulded by different upstream and downstream regulatory elements, which can lead to differences in expression.

It is also important to consider some limitations of the method. The automatic classification of ISH+ or DsRed+ is not flawless: In the ISH case, the punctate and peripheral nature of the labelling makes it difficult to separate cells' signals, which can lead cells adjacent to positive cells to be classified as positive, or to positive cells being missed. In the DsRed case, the labelling is very heterogeneous in terms of brightness, so positive but dim cells are often not considered positive. Both difficulties have probably influenced the estimates.

Another important limitation is that, due to the attenuation of signal with depth in the confocal imaging, information about the ventral parts of hindbrain and midbrain could not be obtained. This is concerning because some of the most relevant neural populations studied in this work are found in these regions, and claims were made here regarding their neurotransmitter identities.

All in all, the fidelity metrics presented here, even if they need to be interpreted carefully, suggest that more care is needed when making claims about neurotransmitter identities based on transgenic lines. While the low sensitivity is not necessarily concerning, increasing mostly the type II error, the relatively low precision is of special concern, as it can lead to erroneous interpretations.

5.3. Oculomotor related activity in the hindbrain

The analysis of functional imaging data presented in this work focused on just two variables: eye position and eye velocity. Motor control of the eyes requires the two types of signals¹⁵, which are thought to be computed in pre-oculomotor circuits, namely in the VSM and the hVPNI^{16,23}. In an electrophysiological study in the goldfish, distinguishing between eye position and eye velocity coding neurons has allowed the identification of two separate regions which possibly correspond to the VSM and the hVPNI²⁴. Yet in zebrafish this position-velocity distinction is not so clear⁵³. As such, understanding eye position and eye velocity coding in the hindbrain is a relevant research direction.

5.3.1. Eye position and eye velocity coding in the larval zebrafish hindbrain

The correlation maps showed that neurons tuned to eye position and eye velocity are found throughout the whole hindbrain and are arranged highly symmetrically, with most neurons responding to ipsiversive eye movements. This is consistent with previous research^{1,8,9}.

In the position map, there was a clear separation between nasal and temporal coding. In the ABN region, both temporal and nasal tuned neurons were found, organized topographically: Neurons located ventrally were mostly temporal tuned while the nasal tuned neurons were found more dorsally. The first group probably corresponds to the ABN motoneurons (which innervate the lateral rectus, responsible for temporal movements) and the second to the ABN interneurons, that connect to the OMN. Consistent with this, temporal tuned neurons are found mostly overlaid with the ABN masks, which were outlined based on cholinergic motoneurons only. The overall image in the ABN region is also quite similar to the data presented by Brysch and colleagues⁵³. In the OMN, neurons were mostly nasal tuned, which is to be expected, since the OMN contains the medium rectus motoneurons, responsible for nasal movement of the eyes. Moreover, the identified neurons are mostly in the dorsal

part of the OMN masks, which is in accordance with the known dorsoventral organization of the OMN motoneurons⁷⁸, further supporting that these are in fact MR motoneurons.

In the velocity map, two things were noteworthy: first, there were more neurons than in the position map. Second, there was no clear spatial separation between nasal and temporal tuned neurons. Both features can perhaps be explained as consequences of the high correlation between eye velocity and stimulus velocity. On the one hand, some of these velocity tuned neurons could be in reality tuned to the stimulus, explaining the increased number of neurons. On the other hand, previous research suggests that binocular integration of optical flow stimuli occurs in the pretectum^{10,57,58}, meaning stimulus-related signals from the pretectum into the hindbrain during the OKR are probably ‘leftward rotation’ and ‘rightward rotation’, instead of ‘left eye temporal’ and ‘right eye nasal’. As such, the absence of any structure in the distribution of nasal and temporal velocity neurons may reflect the fact that stimulus driven neurons labelled as eye velocity tuned fall into just two categories in the hindbrain. Consistent with this hypothesis, Feierstein and colleagues⁹ saw that neurons in the dorsal hindbrain tuned to rotational eye movements fall into just two categories: leftward or rightward rotation.

Another interesting detail in the velocity map is that the region consistent with the hindbrain oscillator (HBO) is visible, and, contrary to previous reports⁷⁹, seems tuned to contraversive movements. Indeed, the checkerboard pattern observed in the HBO and anterior hindbrain (figure 4.6) is the opposite of the one reported by Wolf, Dubreuil and colleagues⁷⁹. One possible explanation for this discrepancy is that these regions were implicated in gaze reorientation through saccades, and the velocity map presented here is of slow eye velocity. In the normal OKR response, saccadic velocity will be negatively correlated with slow eye velocity (because saccades are usually resets, contrary to the direction of the slow movement), so one would expect an opposite map. Thus, the checkerboard areas, which include the HBO and the anterior hindbrain, could be appearing here as a confound and not be truly related to slow eye movements.

5.3.2. Neural populations involved in the OKR

After looking into the spatial organization of eye position and eye velocity correlations, the PVI map was created to separate neurons into eye position only and eye velocity only coding neurons, thus providing a simple metric to cluster neurons into different populations. Notwithstanding the assumptions behind this approach, the PVI map clearly shows a structured distribution of eye position and eye velocity coding neurons. Moreover, if the PVI map is restricted to the ventral hindbrain region, it is quite similar to the one presented by Brysch and colleagues⁵³.

The outlined clusters correspond to both known and previously unknown regions. Known clusters include the ABN region, the OMN, the IO and the HBO. The CDPos has also been previously described, but its function is not clear⁸. The CVmix and MHbPos do not appear to correspond to known neural populations. In general, the clusters had low percentages of either vglut2a+ or gad1b+ neurons. This can be explained with the low sensitivity of the neurotransmitter identification method (62%) which compounds with the low sensitivity of the vglut2a line (62%), meaning that possibly only 40% of actual glutamatergic cells are being classified as such.

In the following discussion, each of the outlined populations is considered. Note that the terms GABAergic and glutamatergic employed below should be interpreted as putative attributions, considering what was discussed in section 5.2.

Motor nuclei related clusters: the ABN region and the OMN

The ABN region is defined here as a large eye position coding region which includes the ABN masks and neurons extending caudally beyond it. About a fifth of neurons in this cluster were considered glutamatergic, a part of which might correspond to ABN interneurons, which are known to be glutamatergic⁸⁰. The OMN cluster was restricted to the motor nucleus and had 5-10% of glutamatergic and GABAergic neurons. Considering the uncertainty associated with neurotransmitter identification, these values are probably not relevant. Yet it is the only cluster in which both vglut2a+ and gad1b+ neurons were identified.

The IO

The IO has long been known to work as an error detector, which works in concert with the cerebellum to promote motor adaptation¹³. Previous studies in zebrafish showed the IO to be active during optokinetic stimulation^{8,53}. In terms of neurotransmitter identity, the IO had around 60% of neurons classified as glutamatergic. This agrees with previous studies in rodents and fish that found the IO neurons, namely the climbing fibres, to be glutamatergic⁸¹⁻⁸³.

The HBO

The HBO has been previously implicated in gaze orientation⁷⁹. As was argued above, the appearance of this cluster could be a confound due to the anti-correlation between saccades and eye velocity. Regarding neurotransmitter identity, according to previous research, the HBO is composed of two stripes, a glutamatergic medial stripe and a GABAergic lateral stripe⁷⁵. This contradicts the data presented here, with no glutamatergic neurons being found in this cluster, even if an appreciable number of GABAergic neurons was found. In fact, the glutamatergic/GABAergic stripe pattern was visible in this area, but the identified neurons did not fall in the glutamatergic stripe, which was quite faint. This could be just because the labelling was weaker, or it could also be that neurons in the HBO responding in this context are different than the neurons responding during the exploratory locomotion studied by Dunn, Mu and colleagues⁷⁵.

Unknown clusters: CDPoS, MHbPos and CVmix

The CDPoS was only previously described by Portugues and colleagues⁸, also as a position-coding region, and the authors suggested it could be related or a part of the hVPNI. Besides confirming the existence of that cluster, the results presented here have also shown that it has a sizeable GABAergic component.

Regarding the two last clusters, it is not clear what they are or what they do. Their locations do not correspond clearly to any previously delineated anatomical structure. Position-coding neurons in the area described here as MHbPos have been found by Brysch and colleagues⁵³ as well, but the authors did not focus on their role, having just speculated that these neurons might carry efference copies of eye position signals. The CVmix is near the brain boundary and could correspond to the lateral reticular nucleus, but it is not clear if it does. No conclusions could be taken about their neurotransmitter identity either. More information is needed to be able to understand the role these clusters might play in the OKR.

5.3.3. Location of the hVPNI and VSM

Considering what is known about the control of the oculomotor system, discussed in detail in section 2.1, what can be said regarding the identified neuronal populations? The ABN, the OMN and the IO have been found in this study, along with unknown populations, but what about two crucial neural populations behind oculomotor control, the hVPNI and VSM?

The hVPNI has been located in rhombomere 7/8 by a combination of functional experiments and optogenetics⁴⁴. Based on correlations with eye position, several authors have identified putative integrator neurons mostly in the ventral hindbrain^{11,12,44,48,84}. By contrast, Portugues and colleagues⁸ suggested the CDpos could also be related to the hVPNI, despite its dorsal position. Brysch and colleagues⁵³ claimed that a position-velocity coding gradient existed caudally and dorsally near the IO, and that the position neurons found here could correspond to integrator neurons. While no definite claims can be made based on the results of this work, the imaged area includes both areas mentioned above, and indeed position coding neurons were found both in the dorsal rhombomere 7/8, in the CDpos, and ventrally, near the IO. The group of position neurons just dorsal and caudal to the IO (see figure 4.7, sagittal view) could correspond to the postulated position-velocity gradient. But it is also true that many other position neurons were found – Finding neurons tuned to eye position is probably insufficient to suppose an integrator role.

Regarding the VSM, it is even less clear where it might be located. The existence of a VSM in larval zebrafish is controversial^{26,85}. Brysch and colleagues⁵³ claimed that the velocity neurons in the position-velocity gradient they observed in rhombomere 7/8 would correspond to VSM neurons. This claim was based on the work of Pastor and colleagues²⁴ which established the location of the hVPNI and VSM as adjacent but separate nuclei in the adult goldfish. The observed gradient in larval zebrafish would simply reflect the immaturity of the circuits⁵³. Considering that the VSM in mammals has been usually located in the vestibular nuclei^{1,86}, which in zebrafish is located caudally and dorsally to the Mauthner cell (location of the medial vestibular nucleus according to the Z-brain atlas⁷¹), this location would not make much sense. In this work no prominent group of eye velocity neurons was found near the Mauthner cell. No further claims can be made on the possible location of this structure.

In conclusion, the definite identification of the hVPNI and the VSM has remained elusive. Although finding neurons coding for eye velocity and eye position is a logical starting point to identify these structures^{24,53}, the observed widespread distributions of both types of neurons made this difficult. Other approaches will be required to be able to accurately locate these oculomotor control structures.

5.4. Conclusion

The results presented above expand on previous work regarding the hindbrain areas involved in oculomotor behaviour and help establishing a methodology for complementing functional imaging experiments with the identification of neurons' neurotransmitter identity.

In particular, the limitations of the use of transgenic lines as reporters of neurotransmitter identity were made evident as well as the need for their validation, and methods for processing and analysing functional imaging data were proposed that can be useful in future research. Understanding the neural circuits underlying the OKR remains a difficult challenge, namely concerning the location of the hVPNI or the VSM, but previous findings regarding the ABN, OMN and the distribution of

oculomotor neurons^{8,9,53} were confirmed. The identification of the neurotransmitter composition of the identified clusters was not highly conclusive but nonetheless serves as an example of the kind of results that can be achieved with this approach.

Future work should focus on more detailed studies of the neural correlates of eye movement control, as well as on the development of better strategies to assign neurotransmitter identity.

6. Bibliography

1. Goldberg, M. E. & Walker, M. F. The Control of Gaze. in *Principles of Neural Science* (eds Kandel, E. R., Koester, J. D., Mack, S. H. & Siegelbaum, S. A.) (McGraw Hill, New York, NY, 2021).
2. Land, M. Eye movements in man and other animals. *Vision Res.* **162**, 1–7 (2019).
3. Raphan, T. & Cohen, B. Brainstem Mechanisms for Rapid and Slow Eye Movements. *Annu. Rev. Physiol.* **40**, 527–552 (1978).
4. Fuchs, A. F., Kaneko, C. R. S. & Scudder, C. A. Brainstem Control of Saccadic Eye Movements. *Annu. Rev. Neurosci.* **8**, 307–337 (1985).
5. Renninger, S. L. & Orger, M. B. Two-photon imaging of neural population activity in zebrafish. *Methods* **62**, 255–267 (2013).
6. Orger, M. B. & Polavieja, G. G. de. Zebrafish Behavior: Opportunities and Challenges. *Annu. Rev. Neurosci.* **40**, 125–147 (2017).
7. Orger, M. B. The Cellular Organization of Zebrafish Visuomotor Circuits. *Curr. Biol.* **26**, R377–R385 (2016).
8. Portugues, R., Feierstein, C. E., Engert, F. & Orger, M. B. Whole-Brain Activity Maps Reveal Stereotyped, Distributed Networks for Visuomotor Behavior. *Neuron* **81**, 1328–1343 (2014).
9. Feierstein, C. E. *et al.* Dimensionality reduction reveals separate translation and rotation populations in the zebrafish hindbrain. *Curr. Biol.* **33**, 3911–3925.e6 (2023).
10. Kubo, F. *et al.* Functional Architecture of an Optic Flow-Responsive Area that Drives Horizontal Eye Movements in Zebrafish. *Neuron* **81**, 1344–1359 (2014).
11. Miri, A. *et al.* Spatial gradients and multidimensional dynamics in a neural integrator circuit. *Nat. Neurosci.* **14**, 1150–1159 (2011).
12. Lee, M. M., Arrenberg, A. B. & Aksay, E. R. F. A Structural and Genotypic Scaffold Underlying Temporal Integration. *J. Neurosci.* **35**, 7903–7920 (2015).
13. Miles, F. A. & Evarts, E. V. Concepts of Motor Organization. *Annu. Rev. Psychol.* **30**, 327–362 (1979).
14. Wolpert, D. M. & Bastian, A. J. Principles of Sensorimotor Control. in *Principles of Neural Science* (eds Kandel, E. R., Koester, J. D., Mack, S. H. & Siegelbaum, S. A.) (McGraw Hill, New York, NY, 2021).
15. Robinson, D. A. The Use of Control Systems Analysis in the Neurophysiology of Eye Movements. *Annu. Rev. Neurosci.* **4**, 463–503 (1981).
16. Cohen, B., Matsuo, V. & Raphan, T. Quantitative analysis of the velocity characteristics of optokinetic nystagmus and optokinetic after-nystagmus. *J. Physiol.* **270**, 321–344 (1977).
17. Cannon, S. C. & Robinson, D. A. Loss of the neural integrator of the oculomotor system from brain stem lesions in monkey. *J. Neurophysiol.* **57**, 1383–1409 (1987).
18. Land, M. F. Eye movements of vertebrates and their relation to eye form and function. *J. Comp. Physiol. A* **201**, 195–214 (2015).
19. Waespe, W. & Henn, V. Neuronal activity in the vestibular nuclei of the alert monkey during vestibular and optokinetic stimulation. *Exp. Brain Res.* **27**, 523–538 (1977).
20. Mustari, M. J. & Fuchs, A. F. Discharge patterns of neurons in the pretectal nucleus of the optic tract (NOT) in the behaving primate. *J. Neurophysiol.* **64**, 77–90 (1990).
21. Jones, G. M. & Milsum, J. H. Characteristics of neural transmission from the semicircular canal to the vestibular nuclei of cats. *J. Physiol.* **209**, 295–316 (1970).

22. Fukushima, K., Kaneko, C. R. S. & Fuchs, A. F. The neuronal substrate of integration in the oculomotor system. *Prog. Neurobiol.* **39**, 609–639 (1992).
23. Robinson, D. A. Integrating with Neurons. *Annu. Rev. Neurosci.* **12**, 33–45 (1989).
24. Pastor, A. M., De la Cruz, R. R. & Baker, R. Eye position and eye velocity integrators reside in separate brainstem nuclei. *Proc. Natl. Acad. Sci.* **91**, 807–811 (1994).
25. Hess, B. J. M. Vestibular response. in *The Oxford Handbook of Eye Movements* (eds Liversedge, S. P., Gilchrist, I. & Everling, S.) 0 (Oxford University Press, 2011). doi:10.1093/oxfordhb/9780199539789.013.0003.
26. Chen, C.-C. *et al.* Velocity storage mechanism in zebrafish larvae. *J. Physiol.* **592**, 203–214 (2014).
27. Laurens, J. & Angelaki, D. E. The functional significance of velocity storage and its dependence on gravity. *Exp. Brain Res.* **210**, 407–422 (2011).
28. Whiteley, A. R. *et al.* Population genomics of wild and laboratory zebrafish (*Danio rerio*). *Mol. Ecol.* **20**, 4259–4276 (2011).
29. Engeszer, R. E., Patterson, L. B., Rao, A. A. & Parichy, D. M. Zebrafish in The Wild: A Review of Natural History And New Notes from The Field. *Zebrafish* **4**, 21–40 (2007).
30. Tavares, B. & Lopes, S. S. The Importance of Zebrafish in Biomedical Research. *Acta Médica Port.* **26**, 583–592 (2013).
31. Teame, T. *et al.* The use of zebrafish (*Danio rerio*) as biomedical models. *Anim. Front.* **9**, 68–77 (2019).
32. Friedrich, R. W., Jacobson, G. A. & Zhu, P. Circuit Neuroscience in Zebrafish. *Curr. Biol.* **20**, R371–R381 (2010).
33. Lister, J. A., Robertson, C. P., Lepage, T., Johnson, S. L. & Raible, D. W. nacre encodes a zebrafish microphthalmia-related protein that regulates neural-crest-derived pigment cell fate. *Development* **126**, 3757–3767 (1999).
34. Svara, F. *et al.* Automated synapse-level reconstruction of neural circuits in the larval zebrafish brain. *Nat. Methods* **19**, 1357–1366 (2022).
35. Ahrens, M. B., Orger, M. B., Robson, D. N., Li, J. M. & Keller, P. J. Whole-brain functional imaging at cellular resolution using light-sheet microscopy. *Nat. Methods* **10**, 413–420 (2013).
36. Pansera, L. *et al.* Zebrafish as an Integrative Model for Central Nervous System Research: Current Advances and Translational Perspectives. *Life* **15**, 1751 (2025).
37. Higashijima, S.-I., Mandel, G. & Fetcho, J. R. Distribution of prospective glutamatergic, glycinergic, and GABAergic neurons in embryonic and larval zebrafish. *J. Comp. Neurol.* **480**, 1–18 (2004).
38. Barreiro-Iglesias, A. *et al.* Distribution of glycinergic neurons in the brain of glycine transporter-2 transgenic Tg(glyt2:Gfp) adult zebrafish: Relationship to brain–spinal descending systems. *J. Comp. Neurol.* **521**, 389–425 (2013).
39. Kinkhabwala, A. *et al.* A structural and functional ground plan for neurons in the hindbrain of zebrafish. *Proc. Natl. Acad. Sci.* **108**, 1164–1169 (2011).
40. Fero, K., Yokogawa, T. & Burgess, H. A. The Behavioral Repertoire of Larval Zebrafish. in *Zebrafish Models in Neurobehavioral Research* (eds Kalueff, A. V. & Cachat, J. M.) 249–291 (Humana Press, Totowa, NJ, 2011). doi:10.1007/978-1-60761-922-2_12.
41. Tabor, K. M. *et al.* Brain-wide cellular resolution imaging of Cre transgenic zebrafish lines for functional circuit-mapping. *eLife* **8**, e42687 (2019).
42. Pang, M. *et al.* Application of the Zebrafish Model in Neurobiological Research. in *Zebrafish Model in Medical Research* (IntechOpen, 2025). doi:10.5772/intechopen.1011592.
43. Denk, W., Strickler, J. H. & Webb, W. W. Two-Photon Laser Scanning Fluorescence Microscopy. *Science* **248**, 73–76 (1990).

44. Miri, A., Daie, K., Burdine, R. D., Aksay, E. & Tank, D. W. Regression-Based Identification of Behavior-Encoding Neurons During Large-Scale Optical Imaging of Neural Activity at Cellular Resolution. *J. Neurophysiol.* **105**, 964–980 (2011).
45. Visualization and analysis of neural activity|IINO Laboratory. <http://molecular-ethology.bs.s.u-tokyo.ac.jp/labHP/E/EResearch/11.html>.
46. Multiphoton Microscopy. *Nikon's MicroscopyU* <https://www.microscopyu.com/techniques/multi-photon/multiphoton-microscopy>.
47. Gonçalves, P. J., Arrenberg, A. B., Hablitzel, B., Baier, H. & Machens, C. K. Optogenetic perturbations reveal the dynamics of an oculomotor integrator. *Front. Neural Circuits* **8**, (2014).
48. Daie, K., Goldman, M. S. & Aksay, E. R. F. Spatial Patterns of Persistent Neural Activity Vary with the Behavioral Context of Short-Term Memory. *Neuron* **85**, 847–860 (2015).
49. Sparks, D. L. The brainstem control of saccadic eye movements. *Nat. Rev. Neurosci.* **3**, 952–964 (2002).
50. Leyden, C., Brysch, C. & Arrenberg, A. B. A distributed saccade-associated network encodes high velocity conjugate and monocular eye movements in the zebrafish hindbrain. *Sci. Rep.* **11**, 12644 (2021).
51. Hering, E. *The Theory of Binocular Vision: Ewald Hering (1868)*. (Springer New York, NY, 1977). doi:<https://doi.org/10.1007/978-1-4613-4148-2>.
52. von Helmholtz, H. *Helmholtz's Treatise on Physiological Optics (Trans. from the 3rd German Ed.)*. (Optical Society of America, 1924). doi:<https://doi.org/10.1037/13536-000>.
53. Brysch, C., Leyden, C. & Arrenberg, A. B. Functional architecture underlying binocular coordination of eye position and velocity in the larval zebrafish hindbrain. *BMC Biol.* **17**, 110 (2019).
54. Dowell, C. K., Hawkins, T. & Bianco, I. H. Subsets of extraocular motoneurons produce kinematically distinct saccades during hunting and exploration. *Curr. Biol.* **35**, 554-573.e6 (2025).
55. Dowell, C. K., Lau, J. Y. N., Antinucci, P. & Bianco, I. H. Kinematically distinct saccades are used in a context-dependent manner by larval zebrafish. *Curr. Biol.* **34**, 4382-4396.e5 (2024).
56. Cochran, S. L., Dieringer, N. & Precht, W. Basic optokinetic-ocular reflex pathways in the frog. *J. Neurosci.* **4**, 43–57 (1984).
57. Wang, K., Hinz, J., Haikala, V., Reiff, D. F. & Arrenberg, A. B. Selective processing of all rotational and translational optic flow directions in the zebrafish pretectum and tectum. *BMC Biol.* **17**, 29 (2019).
58. Naumann, E. A. *et al.* From Whole-Brain Data to Functional Circuit Models: The Zebrafish Optomotor Response. *Cell* **167**, 947-960.e20 (2016).
59. Saxena, S. & Cunningham, J. P. Towards the neural population doctrine. *Curr. Opin. Neurobiol.* **55**, 103–111 (2019).
60. Choi, H. M. T. *et al.* Third-generation in situ hybridization chain reaction: multiplexed, quantitative, sensitive, versatile, robust. *Development* **145**, (2018).
61. Shainer, I. *et al.* A single-cell resolution gene expression atlas of the larval zebrafish brain. *Sci. Adv.* **9**, eade9909 (2023).
62. Marques, I. P. Quantitative Characterization of a zebrafish reporter lines targeting different neurotransmitter populations. (Universidade de Lisboa, 2026).
63. Schindelin, J. *et al.* Fiji: an open-source platform for biological-image analysis. *Nat. Methods* **9**, 676–682 (2012).
64. Stringer, C., Wang, T., Michaelos, M. & Pachitariu, M. Cellpose: a generalist algorithm for cellular segmentation. *Nat. Methods* **18**, 100–106 (2021).
65. Pachitariu, M. *et al.* Suite2p: beyond 10,000 neurons with standard two-photon microscopy. 061507 Preprint at <https://doi.org/10.1101/061507> (2017).

66. Schmidt, U., Weigert, M., Broaddus, C. & Myers, G. Cell Detection with Star-Convex Polygons. in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018* (eds Frangi, A. F., Schnabel, J. A., Davatzikos, C., Alberola-López, C. & Fichtinger, G.) 265–273 (Springer International Publishing, Cham, 2018). doi:10.1007/978-3-030-00934-2_30.
67. Avants, B. B., Epstein, C. L., Grossman, M. & Gee, J. C. Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain. *Med. Image Anal.* **12**, 26–41 (2008).
68. Hunter, J. D. Matplotlib: A 2D Graphics Environment. *Comput. Sci. Eng.* **9**, 90–95 (2007).
69. Cohen, J., Cohen, P., West, S. G. & Aiken, L. S. Multiple Regression/Correlation With Two or More Independent Variables. in *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences* (Routledge, 2003).
70. Stoffel, M. A., Nakagawa, S. & Schielzeth, H. partR2: partitioning R² in generalized linear mixed models. *PeerJ* **9**, e11414 (2021).
71. Randlett, O. *et al.* Whole-brain activity mapping onto a zebrafish brain atlas. *Nat. Methods* **12**, 1039–1046 (2015).
72. Ahrens, M. B., Orger, M. B., Robson, D. N., Li, J. M. & Keller, P. J. Whole-brain functional imaging at cellular resolution using light-sheet microscopy. *Nat. Methods* **10**, 413–420 (2013).
73. Satou, C. *et al.* Transgenic tools to characterize neuronal properties of discrete populations of zebrafish neurons. *Development* **140**, 3927–3931 (2013).
74. Satou, C., Kimura, Y. & Higashijima, S. Generation of Multiple Classes of V0 Neurons in Zebrafish Spinal Cord: Progenitor Heterogeneity and Temporal Control of Neuronal Diversity. *J. Neurosci.* **32**, 1771–1783 (2012).
75. Dunn, T. W. *et al.* Brain-wide mapping of neural activity controlling zebrafish exploratory locomotion. *eLife* **5**, e12741 (2016).
76. Petkova, M. D. *et al.* A connectomic resource for neural cataloguing and circuit dissection of the larval zebrafish brain. 2025.06.10.658982 Preprint at <https://doi.org/10.1101/2025.06.10.658982> (2025).
77. Mogi, K., Tomita, H., Yoshihara, M., Kajiyama, H. & Hara, A. Advances in bacterial artificial chromosome (BAC) transgenic mice for gene analysis and disease research. *Gene* **934**, 149014 (2025).
78. Greaney, M. R., Privorotskiy, A. E., D’Elia, K. P. & Schoppik, D. Extraocular motoneuron pools develop along a dorsoventral axis in zebrafish, *Danio rerio*. *J. Comp. Neurol.* **525**, 65–78 (2017).
79. Wolf, S. *et al.* Sensorimotor computation underlying phototaxis in zebrafish. *Nat. Commun.* **8**, 651 (2017).
80. Mayadali, Ü. S., Fleuriet, J., Mustari, M., Straka, H. & Horn, A. K. E. Transmitter and ion channel profiles of neurons in the primate abducens and trochlear nuclei. *Brain Struct. Funct.* **226**, 2125–2151 (2021).
81. Yu, Y., Fu, Y. & Watson, C. The Inferior Olive of the C57BL/6J Mouse: A Chemoarchitectonic Study. *Anat. Rec.* **297**, 289–300 (2014).
82. Grandes, P., Ortega, F. & Streit, P. Glutamate-immunoreactive climbing fibres in the cerebellar cortex of the rat. *Histochemistry* **101**, 427–437 (1994).
83. Ikenaga, T., Morita, S. & Finger, T. E. Histological and Molecular Characterization of the Inferior Olivary Nucleus and Climbing Fibers in the Goldfish, *Carassius auratus*. *Zoolog. Sci.* **40**, 141–150 (2023).
84. Vishwanathan, A. *et al.* Electron Microscopic Reconstruction of Functionally Identified Cells in a Neural Integrator. *Curr. Biol.* **27**, 2137–2147.e3 (2017).

85. Beck, J. C., Gilland, E., Tank, D. W. & Baker, R. Quantifying the Ontogeny of Optokinetic and Vestibuloocular Behaviors in Zebrafish, Medaka, and Goldfish. *J. Neurophysiol.* **92**, 3546–3561 (2004).
86. Yokota, J.-I., Reisine, H. & Cohen, B. Nystagmus induced by electrical stimulation of the vestibular and prepositus hypoglossi nuclei in the monkey: evidence for site of induction of velocity storage. *Exp. Brain Res.* **92**, 123–138 (1992).
87. Ángeles Fernández-Gil, M., Palacios-Bote, R., Leo-Barahona, M. & Mora-Encinas, J. P. Anatomy of the Brainstem: A Gaze Into the Stem of Life. *Semin. Ultrasound CT MRI* **31**, 196–219 (2010).
88. Ingle, D. J. Optic Tectum. in *Comparative Neuroscience and Neurobiology* (ed. Irwin, L. N.) 100–101 (Birkhäuser, Boston, MA, 1988). doi:10.1007/978-1-4899-6776-3_40.
89. optic tectum. *Zebrafish UCL* <http://zebrafishucl.org/midbrain-regions/optic-tectum> (2019).
90. Moens, C. B. & Prince, V. E. Constructing the hindbrain: Insights from the zebrafish. *Dev. Dyn.* **224**, 1–17 (2002).
91. Ahrens, M. B. *et al.* Brain-wide neuronal dynamics during motor adaptation in zebrafish. *Nature* **485**, 471–477 (2012).
92. Orger, M. B., Kampff, A. R., Severi, K. E., Bollmann, J. H. & Engert, F. Control of visually guided behavior by distinct populations of spinal projection neurons. *Nat. Neurosci.* **11**, 327–333 (2008).
93. Asakawa, K. & Kawakami, K. Protocadherin-Mediated Cell Repulsion Controls the Central Topography and Efferent Projections of the Abducens Nucleus. *Cell Rep.* **24**, 1562–1572 (2018).
94. Félix, R. *et al.* Structural and Functional Organization of Visual Responses in the Inferior Olive of Larval Zebrafish. *J. Neurosci.* **44**, (2024).
95. Krauzlis, R. J., Goffart, L. & Hafed, Z. M. Neuronal control of fixation and fixational eye movements. *Philos. Trans. R. Soc. B Biol. Sci.* **372**, 20160205 (2017).
96. Scudder, C. A., Kaneko, C. R. & Fuchs, A. F. The brainstem burst generator for saccadic eye movements. *Exp. Brain Res.* **142**, 439–462 (2002).
97. McFarland, J. L. & Fuchs, A. F. Discharge patterns in nucleus prepositus hypoglossi and adjacent medial vestibular nucleus during horizontal eye movement in behaving macaques. *J. Neurophysiol.* **68**, 319–332 (1992).
98. de NÓ, R. L. VESTIBULO-OCULAR REFLEX ARC. *Arch. Neurol. Psychiatry* **30**, 245–291 (1933).
99. Ito, M. Neural design of the cerebellar motor control system. *Brain Res.* **40**, 81–84 (1972).
100. Zubricky, R. D. & Das, J. M. Neuroanatomy, Superior Colliculus. in *StatPearls* (StatPearls Publishing, Treasure Island (FL), 2026).
101. Simpson, J. I. The Accessory Optic System. *Annu. Rev. Neurosci.* **7**, 13–41 (1984).
102. Giolli, R. A., Blanks, R. H. I. & Liu, F. The accessory optic system: basic organization with an update on connectivity, neurochemistry, and function. in *Progress in Brain Research* vol. 151 407–440 (Elsevier, 2006).
103. Masseck, O. A. & Hoffmann, K.-P. Comparative Neurobiology of the Optokinetic Reflex. *Ann. N. Y. Acad. Sci.* **1164**, 430–439 (2009).
104. Klar, M. & Hoffmann, K.-P. Visual direction-selective neurons in the pretectum of the rainbow trout. *Brain Res. Bull.* **57**, 431–433 (2002).
105. Masseck, O. A. & Hoffmann, K.-P. Question of Reference Frames: Visual Direction-Selective Neurons in the Accessory Optic System of Goldfish. *J. Neurophysiol.* **102**, 2781–2789 (2009).
106. Schoonheim, P. J., Arrenberg, A. B., Bene, F. D. & Baier, H. Optogenetic Localization and Genetic Perturbation of Saccade-Generating Neurons in Zebrafish. *J. Neurosci.* **30**, 7111–7120 (2010).

107. Martins, S. *et al.* Toward an Integrated Zebrafish Health Management Program Supporting Cancer and Neuroscience Research. *Zebrafish* **13**, S-47 (2016).
108. Vladimirov, N. *et al.* Light-sheet functional imaging in fictively behaving zebrafish. *Nat. Methods* **11**, 883–884 (2014).
109. Förster, D. *et al.* Genetic targeting and anatomical registration of neuronal populations in the zebrafish brain with a new set of BAC transgenic tools. *Sci. Rep.* **7**, 5230 (2017).
110. Hörl, D. *et al.* BigStitcher: reconstructing high-resolution image datasets of cleared and expanded samples. *Nat. Methods* **16**, 870–874 (2019).
111. Stringer, C. & Pachitariu, M. Computational processing of neural recordings from calcium imaging data. *Curr. Opin. Neurobiol.* **55**, 22–31 (2019).
112. Giovannucci, A. *et al.* CaImAn an open source tool for scalable calcium imaging data analysis. *eLife* **8**, e38173 (2019).
113. Ramirez, A. D. & Aksay, E. R. F. Ramp-to-threshold dynamics in a hindbrain population controls the timing of spontaneous saccades. *Nat. Commun.* **12**, 4145 (2021).
114. Tanaka, R. & Portugues, R. Algorithmic dissection of optic flow memory in larval zebrafish. *Curr. Biol.* **35**, 4870-4881.e4 (2025).
115. Walt, S. van der *et al.* scikit-image: image processing in Python. *PeerJ* **2**, e453 (2014).
116. Virtanen, P. *et al.* SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* **17**, 261–272 (2020).
117. Arzt, M. *et al.* LABKIT: Labeling and Segmentation Toolkit for Big Image Data. *Front. Comput. Sci.* **4**, (2022).
118. Caicedo, J. C. *et al.* Nucleus segmentation across imaging experiments: the 2018 Data Science Bowl. *Nat. Methods* **16**, 1247–1253 (2019).
119. Rohlfing, T. & Maurer, C. R. Nonrigid image registration in shared-memory multiprocessor environments with application to brains, breasts, and bees. *IEEE Trans. Inf. Technol. Biomed.* **7**, 16–25 (2003).

Appendix A - Comparative neuroanatomy of eye movement control

A.1 General anatomy of the brainstem

The control pathways underlying (horizontal) saccadic, vestibular and optokinetic movements are all located in the brainstem. As such, a brief anatomical overview of the brainstem is presented here.

The brainstem is caudal to the diencephalon (which includes the thalamus and hypothalamus) and rostral to the spinal cord, and it is divided along the rostro-caudal axis between the midbrain, the pons and the medulla oblongata. It also has a dorsoventral organization in three parts, the tectum (more dorsal), the tegmentum, and the basis (more ventral)⁸⁷. The ABN and OMN are located in the brainstem, in the pons and midbrain, respectively.

In larval zebrafish, most of the midbrain is occupied by the optic tectum, a structure homologous to the mammalian superior colliculus. In non-mammalian vertebrates like fish, amphibians and birds, the optic tectum is the fundamental visual processing centre⁸⁸: It receives retinal projections from the contralateral eye, organized in a topographical way, thereby forming a retinotopic map, a paradigmatic feature of visual processing⁸⁹. In the tegmentum, the oculomotor nucleus can be found. The hindbrain includes the cerebellum and the medulla oblongata and is involved in the control of movement and locomotion⁷⁵. In early developmental stages, the hindbrain is divided into 7 or 8 segments called rhombomeres⁹⁰. In the larva, the cerebellum is located on rhombomere 1 and is involved in motor adaptation⁹¹. The remaining rhombomeres form the medulla oblongata, where there are several structures involved in motor control, namely, the reticulospinal neurons, which control movements of the tail⁹², the abducens nuclei, which are involved in eye movements⁹³ and the inferior olive, which acts together with the cerebellum to promote motor adaptation^{91,94}.

A.2 Anatomical localization of oculomotor control structures in mammals

In mammals, saccades can be voluntary, spontaneous, or a part of VOR/OKR movements. While the first type has been the focus of much research, the latter two still require further investigations. Nonetheless, it is known that the main site of interest is the Paramedian Pontine Reticular Formation (PPRF), situated near the ABN^{3,4}. Located in this structure are the ‘Burst’ Neurons (BN) that give a phasic command to promote a saccade. Burst neurons receive tonic inhibition by omnipause neurons, so any upstream saccade-promoting command must directly inhibit them⁹⁵. BNs send their velocity signal to the ABN and to the horizontal VPNI (hVPNI), which supplies the position signal to ABN⁹⁶. The site of the hVPNI has long been a source of interest and has been located in the Nucleus Prepositus Hypoglossi (NPH), a structure close to the vestibular nuclei^{17,97}.

The horizontal VOR circuitry has been classically described as the ‘three-neuron arc’⁹⁸, requiring solely three neurons: a first-order vestibular neuron conveys the head velocity signal from the vestibular system to the Vestibular Nucleus (VN), located in the dorsal medulla and pons, from where a second-order vestibular neuron connects to the ipsilateral ABN, activating or inhibiting motoneurons, according to the direction of the turning. While this simple circuit explains in part the speed of the reflex, it is known that there are other elements, namely the hVPNI, located in NPH, and the VSM, located within the vestibular nucleus itself^{1,86}. The VOR must be continuously adjusted and calibrated to ensure its accuracy. The anatomical correlates of this calibration are well known, as they have formed the basis for more general models of motor adaptation. Namely, the most influential

model of motor adaptation, the Ito model, proposed that the cerebellum is the brain structure where motor learning takes place⁹⁹. The Inferior Olive (IO), a nucleus located in the medulla oblongata, functions as an error detector that connects to the cerebellum via climbing fibres, which can modify the strengths of different synaptic connections within the cerebellum to change motor behaviour appropriately¹³.

The OKR circuitry shares important parts with the VOR, as one would expect from the high complementarity of these responses. The main difference is the source of the velocity signal, which in the case of the OKR is a retinal slip signal. Most visual information coming from the retina travels via the optic nerve and crosses hemispheres in the optic chiasm, entering into the optic tract, from where it will go to the thalamus and then to the visual cortex. However, some axons diverge from the optic tract and connect to different midbrain structures, including the Nucleus of the Optic Tract (NOT), the Accessory Optic System (AOS) (in the pretectum) and the Superior Colliculus (SC) (in the tectum), which have important roles in gaze and postural control¹⁰⁰. Among them are axons of direction-selective retinal ganglion cells, which carry information about whole-field visual motion. In the AOS, which is composed of three terminal nuclei, each nucleus responds to whole-field movement in a different preferred direction. The Dorsal Terminal Nucleus (DTN) is tuned to horizontal whole-field motion and so to the detection of horizontal retinal slip¹⁰¹. The NOT, closely associated with the DTN in terms of both anatomical location and functional role, also responds to horizontal retinal slip. Accordingly, several authors call the structure responsible for relaying the retinal slip signal by the name NOT/DTN¹⁰².

A.3 Anatomical localization of oculomotor control structures in fish

Knowledge about the neuroanatomy of oculomotor control in fish is lacking when compared to mammals and primates. The understanding of circuit arrangement presented before for mammals is still a work in progress for fish, and larval zebrafish specifically. Notwithstanding that, the putative homologs to the NPH, PPRF and NOT/DTN have been successfully found in larval zebrafish and are presented below.

Pioneering studies of the hVPNI in the goldfish (*Carassius auratus*) paved the way for the study of neural integration in zebrafish. An electrophysiological study by Pastor et al. (1994) located the hVPNI in a brainstem nucleus caudal to the ABN and proximal to the IO, consistent with the anatomical location of the hVPNI in the NPH in mammals²⁴. Moreover, it established there were separate nuclei encoding eye position and velocity, area I and area II, corresponding to the putative hVPNI and VSM, respectively. In larval zebrafish, the hVPNI is located in the most caudal part of the ventral hindbrain, in rhombomeres 7 and 8. In a landmark study⁴⁴, by using functional imaging and linear regression, the authors found a group of cells highly correlated to saccadic velocity and eye position - the expected tuning for neurons in the hVPNI - in this region. Further functional imaging studies showed that within this region there is a velocity-position coding gradient, in contrast to the clear separation of position encoding and velocity encoding neurons in areas I and II in the goldfish, but in accordance with the expected from mammal studies⁵³.

The detection of horizontal optic flow in vertebrates takes place in the pretectum: In mammals, as has been previously discussed, it occurs in the NOT/DTN; in amphibians, reptiles and birds, in a pretectal nucleus called nucleus *lentiformis mesencephali*¹⁰³; and in electrophysiological studies of goldfish and rainbow trout, it has been located as well in the pretectal region, termed *area preterminalis* (APT)^{104,105}. Kubo and colleagues¹⁰ showed that in larval zebrafish it is no different. By photoactivating the pretectum the authors showed robust optokinetic responses in the absence of sensory input, and by photoinhibiting the same area during an optokinetic stimulus, the response was abolished, therefore implicating this area in responses to optic flow.

The saccade-triggering area of the larval zebrafish brain, putatively homologous to the mammalian PPRF, was located in the medial part of rhombomere five using optogenetics¹⁰⁶.

Appendix B - Zebrafish strains and animal care

Adult fish were maintained and reared at the Champalimaud Foundation Fish Facility at 25°C on a 14:10 hour light cycle following standard methods¹⁰⁷. Transgenic lines used in this work are listed in Table B.1. All transgenic lines were in a nacre background (Melanocyte Inducing Transcription Factor A (*mitfa*) $-/-$) to ensure transparency of the larvae. Appropriate crosses were set using the transgenic fish lines listed in Table 3.1 to obtain larvae expressing the desired fluorophores (GCaMP6s, DsRed, mCherry) in the required domains for each experiment (i.e., in the whole brain, in glutamatergic neurons, etc.). Embryos were kept at 28°C in E3 embryo medium (5 mM NaCl, 0.17 mM KCl, 0.33 mM CaCl₂, 0.33 mM MgSO₄ and 3x 10⁻⁵ % Methylene Blue in aquaculture system water) on a 12:12 hour light cycle. The crosses and embryo care were ensured by the fish facility.

At 4-5 days post-fertilization (dpf), fish were screened for fluorescence using a fluorescence stereoscope. For each petri dish with fish larvae, 1 mL of Tricaine 25x (15 mM MS-222, 20mM Tris, pH7) was used to immobilize the fish, which were then individually checked under the stereoscope for fluorescence in the green and red channels. Only fish with signal in both channels were selected for further experiments. These were incubated in petri dishes with about 20 fish/dish in a live feed culture of rotifers and artemia, provided by the fish facility, which was renewed after two days. Larvae which were unused for experiments were euthanised following approved protocols. Sexual differentiation occurs at a later larval stages, therefore the sex of the animals cannot be reported. All experimental procedures were approved by the Champalimaud Foundation Ethics Committee and the Portuguese Direção-Geral de Agricultura e Veterinária and were performed according to the European Directive 2010/63/EU.

Table B.1. Transgenic Lines used in this work^{73,74,108,109}

Transgenic Line	Name in text	Source	ZFIN identifier
Tg(<i>elavl3</i> :Hsa.H2B-GCaMP6s) jf5Tg	<i>elavl3</i> :H2B-GCaMP6s	Ahrens Lab (Vladimirov et al., 2014)	ZDB-ALT-141023-2
TgBAC(<i>slc17a6b</i> :LOXP-DsRed-LOXP-GFP) <i>nns14</i>	<i>vglut2a</i> :DsRed	Higashijima Lab (Satou et al., 2012)	ZDB-ALT-110413-5
TgBAC(<i>gad1b</i> :LoxP-RFP-LoxP-GFP) <i>nns26</i>	<i>gad1b</i> :DsRed	Higashijima Lab (Satou et al., 2013)	ZDB-ALT-131127-7
Tg[<i>chata</i> :GAL4] <i>mpn202</i> ; Tg[UAS-mCherry]	<i>chata</i> mCherry	Baier lab (Förster et al., 2017)	ZDB-ALT-170908-4

Appendix C – Confocal image processing

The confocal image processing pipeline used for the line validation is described in detail below. The development and validation of this pipeline was done elsewhere⁶².

Images were first deconvolved with the AiryScan Processing function in Zeiss' ZEN software (with 3D processing and Auto Filter activated and the strength set to standard), stitched and downsampled to 0.1x0.1x0.25 μm . For the stitching, the FIJI plugin BigStitcher was used¹¹⁰: First, the images were resaved as multi-resolution hdf5 files, using the manual mipmap setup with predefined parameters. Afterwards, the tiles were manually aligned. To refine the alignment, the stitching wizard was used, with the predefined parameters. Finally, images were stitched with the image fusion command.

The green channel, which contained the pan-neuronal nuclear labelling with GCaMP6s, was further processed in FIJI before being fed into CellPose⁶⁴ for nuclei segmentation. The images were background subtracted (with a rolling-ball radius of 50 pixel), 3D median filtered (2 pixel radius), and two methods were used to adjust contrast: a custom-made script to correct for depth-dependent brightness attenuation, and a macro to apply Contrast Limited Adaptive Histogram Equalization (CLAHE) to every slice of the stack, with the following parameters: blocksize =127, 256 bins, max slope =3.

A fine-tuned CellPose model was used to segment nuclei based on the GCaMP6s signal in overlapping blocks of 100 slices. The following parameters were used: --restore type denoise_cyto3; -use_gpu; --do_3D; --anisotropy 2.5; --flow_threshold 0.4; --cellprob_threshold -5; --stitch_threshold 0.4; --niter 600; --min_zise=2000; --flow3D_smooth 3. The resulting masks were then filtered based on size (only masks with a volume between 3000 and 70000 pixels were kept) and a morphological closure was applied to fill any small holes within the masks. A second quality filtering was done based on roundness, with only masks with an average roundness (per 2D slice) of 0.58 being kept. The resulting filtered mask overlapping blocks were then converted to non-overlapping blocks and any masks crossing the boundary of each block were discarded, to prevent pseudo-replication.

The nuclear masks were then applied to the DsRed and alexa647 channels to classify each cell as DsRed positive or negative and ISH positive or negative, based on the fraction of pixels with a signal level above a certain threshold. For the DsRed, the fraction of very bright pixels (above the 85% percentile of pixel intensity values in each slice) was measured in each ROI, and Otsu's thresholding (calculated for the distribution of values of each block) was used to automatically divide cells into DsRed+ and DsRed-. For the ISH, there is an important difference - the ISH labelling is exclusively cytoplasmic, while the masks are nuclear. Therefore, donut-shaped masks were defined by creating a Voronoi tessellation of the brain, which attributed each extra-nuclear pixel to a nearby nuclear mask, and then removing the nucleus. In these masks, the fraction of very bright pixels (above 80% percentile) was measured, and Otsu's thresholding (also calculated per block) was used to automatically classify the cells as ISH+ or ISH-.

Appendix D - Functional image processing

Calcium imaging datasets typically require three processing steps: motion correction, cell segmentation and trace extraction¹¹¹. In this thesis, a python pipeline that relied on open-source tools was assembled to accomplish these steps. Although there are packages dedicated to this task, most notably Suite2p⁶⁵ and CaImAn¹¹², two specific features of this work motivated the use of a more customized pipeline. The first was the pseudo-volumetric imaging approach employed, in contrast to single-plane imaging or fast volumetric imaging that are commonly used^{50,54,72}, and which led to particular needs in the motion correction and segmentation part. The other feature was the goal of single-cell identification of neurotransmitter identity, which required a segmentation approach capable of identifying anatomically correct boundaries for cells. This is why an anatomically based segmentation method was used, instead of activity-based segmentation approaches, which are more commonly used^{18,44,113,114}.

D.1 Motion correction

Open-source motion correction tools for calcium imaging data were developed mainly for single plane data (Suite2P, CaImAn) or for volumetric imaging (CaImAn 3D)^{65,112}. The pseudo-volumetric imaging approach followed here therefore required a custom two-part motion-correction algorithm. First, frames from each plane were aligned to each other using Suite2P's `registration_wrapper` function with parameters `nimg_init: 100`, `two_step_registration: True`, `do_bidiphase: True`, `bidiphase: 0`, `bidi_corrected: False`; and the remainder ones set to their default values. The last three parameters were set so that the line offset caused by bidirectional scanning was also corrected.

In the second part, planes were aligned to each other. To do so, the necessary shift between adjacent planes was computed using the `phase_cross_correlation()` function from `skimage.registration`¹¹⁵. The input images were the average intensity projection of each plane's frames, but it was found that better results were obtained by cropping the field of view to a smaller, centred rectangle within the image (roughly 400x600 pixels). After the shift was calculated, it was applied to every frame of the subsequent plane with the `shift` function from `scipy.ndimage`¹¹⁶. Suite2P's motion correction identifies frames with too much motion as bad frames, however it was found that Suite2P's identified bad frames did not always match with the actual bad frames. As such, a correlation-based approach was used to identify bad frames, in which frames which had a correlation with other frames below two standard-deviations of the average correlation between frames were excluded. The missing frames were then linearly interpolated.

D.2 Nuclei segmentation and trace extraction

After motion correction, the average projection of each plane's 250 frames was segmented with StarDist2D⁶⁶. StarDist was chosen for segmenting neurons for two main reasons: First, it is developed specifically for nuclei segmentation, which is well suited for the acquired images based on nuclear-expressed GCaMP. Second, it is lightweight and easy to access. Namely, it is readily available as a FIJI plugin and does not have high memory and computing requirements. Segmentation with StarDist relied on a custom-trained model in 2D (see section D.3).

The resulting ROIs were then coarsely filtered, removing any ROIs with an area greater than 150 pixels² or with a peak Signal to Noise Ratio (defined as the ratio of the maximum value of the calcium trace over the standard deviation of the calcium trace in frames which were below the 20th percentile of activity) smaller than 1. These parameters were chosen to provide a crude filtering that removed obviously fake cells, such as large blobs in skin regions. Finally, a calcium activity trace was extracted from each ROI as the average intensity within the ROI over the course of the 250 frames of that plane. This completed the 2D neuron segmentation part.

In contrast with the motion-correction and the 2D segmentation, based on widely used open-source tools, the final step of the pipeline was developed in an *ad hoc* manner to go from the anatomically segmented 2D nuclei to 3D neurons. 3D stitching of 2D nuclei is essential because since the z-step of the 2-photon imaging is 1 micron, each cell appears on multiple planes. This is an important source of replication, as in each plane every stimulus is repeated. Correspondingly, the stitching is also crucial to avoid pseudoreplication. In this pipeline, 2D ROIs were stitched together based on a threshold for the maximum distance between centroids, set to 4 pixels, and the Intersection over Union - IoU, defined for any two adjacent ROIs as:

$$(D.1) \quad IoU = \frac{ROI\#1 \cap ROI\#2}{ROI\#1 \cup ROI\#2}$$

set to 0.6. The thresholds were chosen heuristically. If there were multiple candidate ROIs to be stitched to a previous one, the one with the highest score, calculated as the average between centroid distance and the IoU, was chosen. A minimal and maximal z-extent of 3 and 7 planes, respectively, for each 3D stitched ROI was defined to achieve realistic cell dimensions.

It should be noted that, since the stitching was based on heuristics regarding 2D ROI overlap and cell dimensions, 3D ROIs do not correspond necessarily to individual neurons. For example, one 3D ROI could have two slices belonging to the neuron below it. It is difficult to assess the extent to which this might have happened due to the low resolution in the z-axis, which often makes it impossible to individualize neurons visually. Considering the promising results obtained with StarDist2D, a possible future avenue to deal with this issue would be to train a StarDist3D model.

Finally, a trace for each putative neuron in 3D was built by calculating the DF/F of each of its constituent 2D ROI's traces with the formula:

$$(D.2) \quad \frac{DF}{F} = \frac{F - F_{median}}{F_{median}},$$

where F is the trace and F_{median} is its median, and then concatenating these DF/F traces and z-scoring the result.

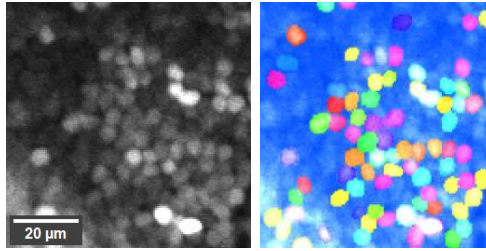
D.3 Training a StarDist2D model

StarDist is a nuclei-segmentation tool designed to detect star-convex objects, i.e., objects in which any radius from the centre to the perimeter of the object's shape is contained within the object⁶⁶. Preliminary experiments with this tool, available as a FIJI plugin, revealed that it performed better than functional-based segmentation approaches. As such, a model was trained to tune StarDist to the specificities of the imaging data acquired in the experiments for this work, in order to improve segmentation quality. An annotated dataset was built that included 24 128x128 pixel images and 2 256x256 pixel images from a functional image experiment (hereby called the 'specific dataset'). The images were chosen so as to form a heterogeneous dataset, covering regions with sparse or dense cell populations, at different depths, in different parts of the hindbrain and with different brightness. These images were manually segmented using the FIJI plugin Labkit¹¹⁷ to achieve single pixel labelling (see Figure C.1A for an example image). The annotated dataset was then merged with the publicly

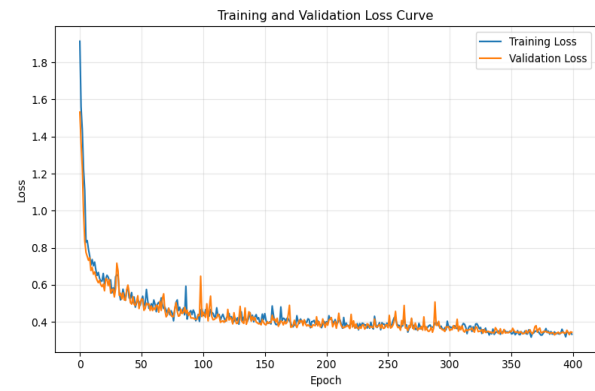
available dataset used to train StarDist's first model, the Kaggle 2018 Data Science Bowl dataset (image set [BBBC038v1](#), ¹¹⁸) (the 'generalist dataset'), creating the final annotated dataset.

Training was done with the python notebooks provided in StarDist's project page (<https://github.com/stardist/stardist/tree/main/examples>), and involved the partition of the dataset between a training and a test dataset, guaranteeing the presence of the 'specific dataset's' images in both. The model trained for 2 hours on a standard computer with GPU and 32Gb RAM. Figure C.1B showcases model performance during the training by evaluating the loss function across epochs for the training and the validation dataset. To further quantify model performance, one needs to establish a criterion to count a cell as a match or as mismatch between the manually and automatically segmented datasets. In this case, an IoU threshold of 0.5 was used, i.e., if the IoU between the annotated and the segmented mask was greater than 0.5, it was considered a match. With this criterion, the trained model achieved a precision of 0.94 (i.e., 94% of segmented cells are actual cells), a sensitivity of 0.88 (i.e., 88% of the actual cells were segmented) and an accuracy of 0.84 (i.e., 84% of the model's predictions matched reality). As can be seen in an example image Figure C.1C, the segmentation quality is very high, although there are occasional mis-segmented nuclei that appear as large blobs in regions of skin or at the border.

(A)



(B)



(C)

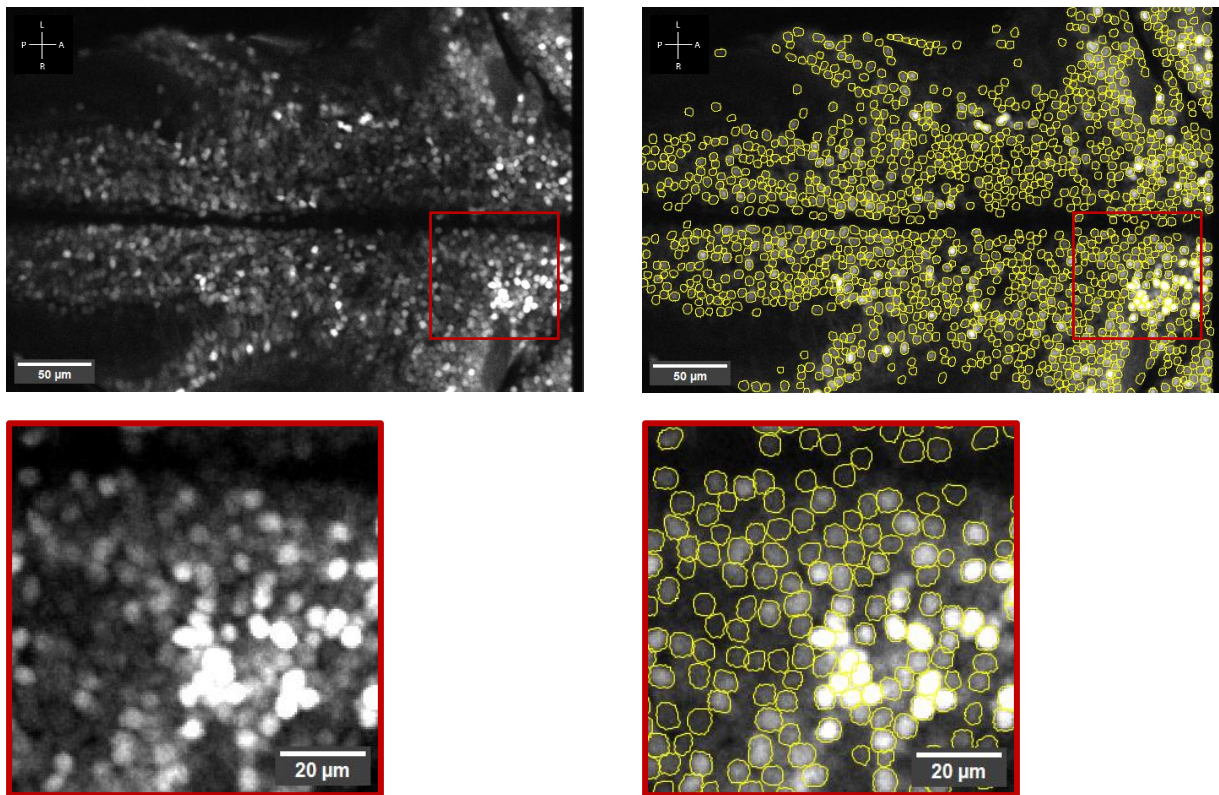


Figure D.1. StarDist model training. (A) example image from the ‘specific dataset’ (left) and its manual annotation in LabKit (right). (B) Loss throughout the model training process for the training (blue) and validation (orange) sets. (C) Example of how the model performs. (Top) An example plane of a fish brain (left) and its identified ROIs with StarDist (right). Red square shows the region zoomed-in below. (Bottom) Zoomed-in region from the plane above.

Appendix E - Building an anatomical reference brain

Several *elavl3:H2B-GCaMP6s* fish were mounted in agarose with Tricaine to be imaged immobilized in the 2-Photon microscope. The mounting protocol is similar to the one described for functional experiments (see section 3.2.1), but fish were first immobilized by immersion in Tricaine 1X, placed on a drop of 2% low melting point agarose in Tricaine 1X, and the eyes and tail were not freed. High-quality, whole-brain anatomical stacks were acquired that spanned a volume of $767 \times 456 \times 260$ micra. The best five ones were picked to proceed with the reference brain building, and the best of those was chosen as the 'seed' brain. Initially, two other images were registered to the first using ANTs, and then averaged, creating a new reference. Then the five images were all registered to this new reference and averaged, updating the reference brain. This step was repeated two times, creating an 'average' fish brain.

After the reference brain was built, the data from each fish from the functional experiments was registered to this reference using the Computational Morphometry Toolkit (CMTK - RRID:SCR_002234¹¹⁹). The images were transformed using the registration, warp and reformatx commands; and the centroid coordinates of the identified 3D neurons were also transformed into the reference space using streamxform.

To facilitate the identification of known hindbrain structures involved in oculomotor behaviour (namely the abducens (ABN) and oculomotor (OMN) nuclei), whole-brain images of *elavl3:H2B-GCaMP6s* x *chata:mCherry* were taken in the 2-photon microscope with the laser tuned to 1020nm and following a protocol identical to the described above (mounting in tricaine, image acquisition, motion correction of the images and registration to the template with CMTK). The *chata* line labels motor neurons (which are cholinergic), and therefore labels neurons in the motor cranial nerve nuclei. Manual drawn masks corresponding to the ABN and the OMN were drawn based on the resulting image stacks (Figure E.1). Note that the OMN and the trochlear nucleus are indistinguishable based on this marker alone, so the OMN mask also includes the trochlear nucleus. The Mauthner cells are another useful landmark to locate anatomical structures in the hindbrain, and their location was visually pinpointed in the reference brain.

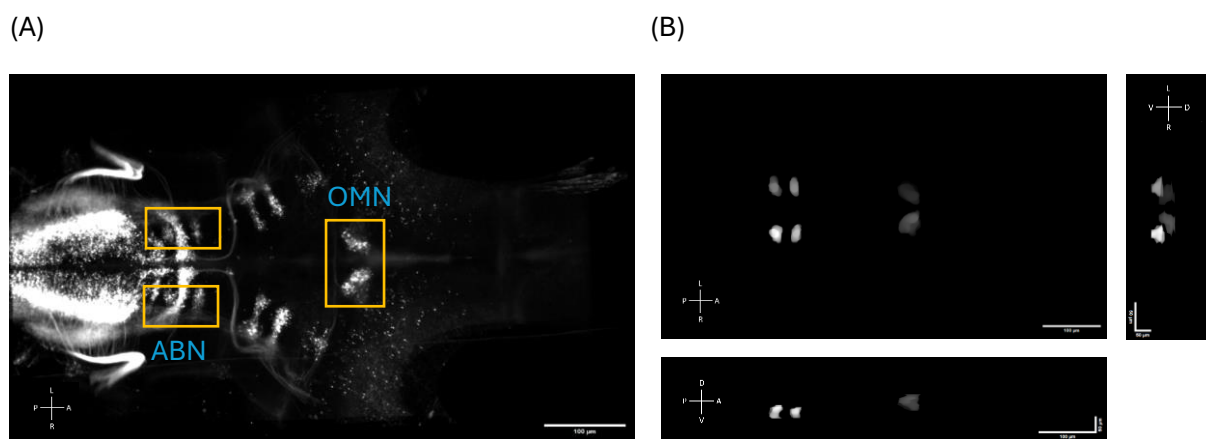
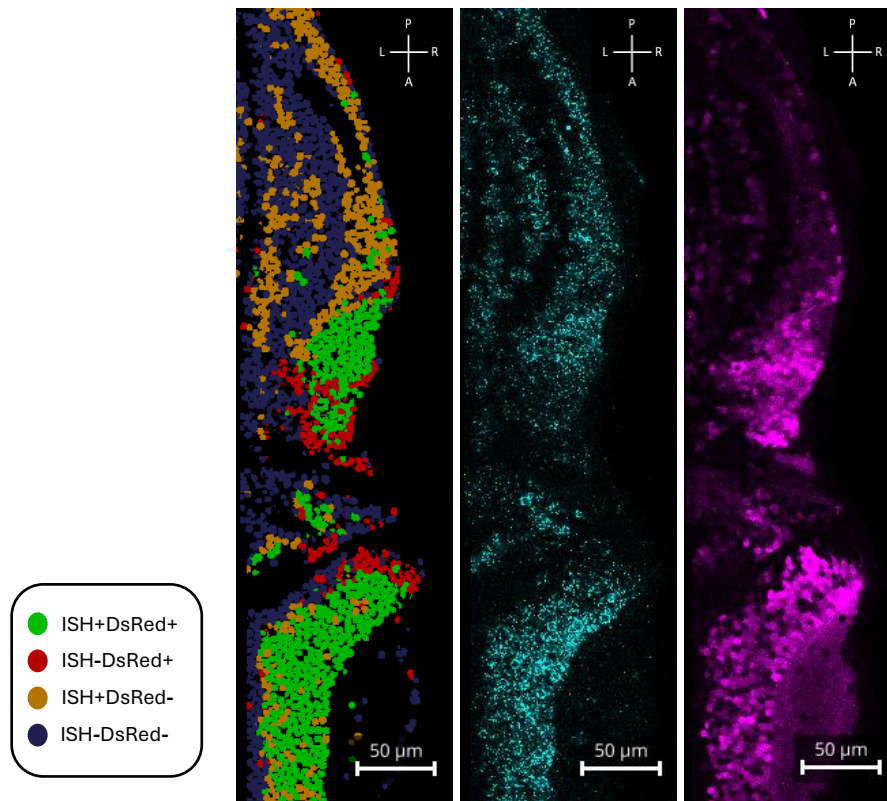


Figure E.1. Determining ABN and OMN location with the *chata:mCherry* line. (A) Standard-deviation projection image of the average of four fish registered to the reference brain. The OMN and ABN are highlighted. Note that the ABN is partially hidden by the more dorsal facial motor nucleus. (B) Resulting hand drawn masks in a horizontal (top), sagittal (bottom) and coronal (right) projection.

Appendix F – Supplementary figures

(A)



(B)

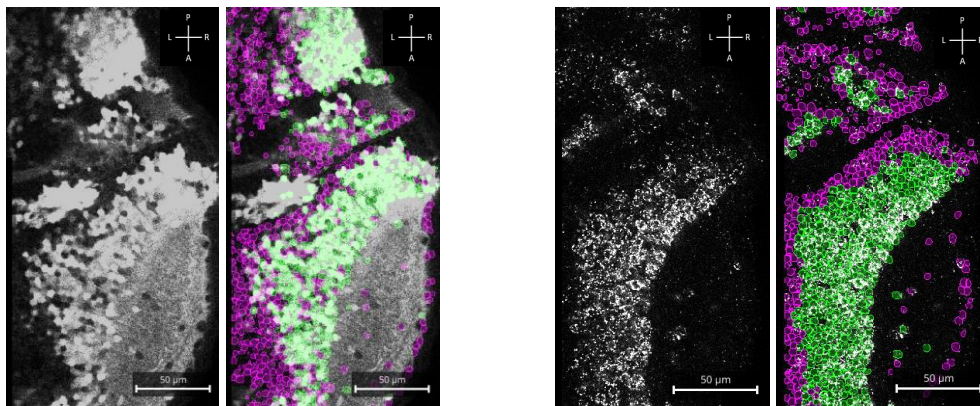


Figure F.1. Performance of the automatic classification of neurons. (A) Final classification of neurons in an example slice of an example fish. (Left) Classification: Green – ISH+ DsRed+, Red – ISH- DsRed+, Orange – ISH+ DsRed-, and Dark purple – ISH- DsRed-. (Middle) ISH labelling. (Right) DsRed labelling. (B) Classification of neurons as DsRed+ (left) and as ISH+ (right). Each pair of images shows the original image (an example slice from an example fish) and in the right the automatic classification with green being positive cells and magenta being negative ones.

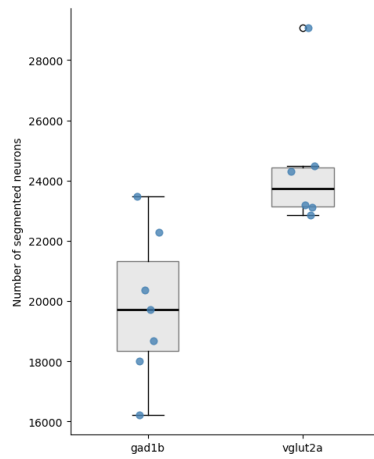
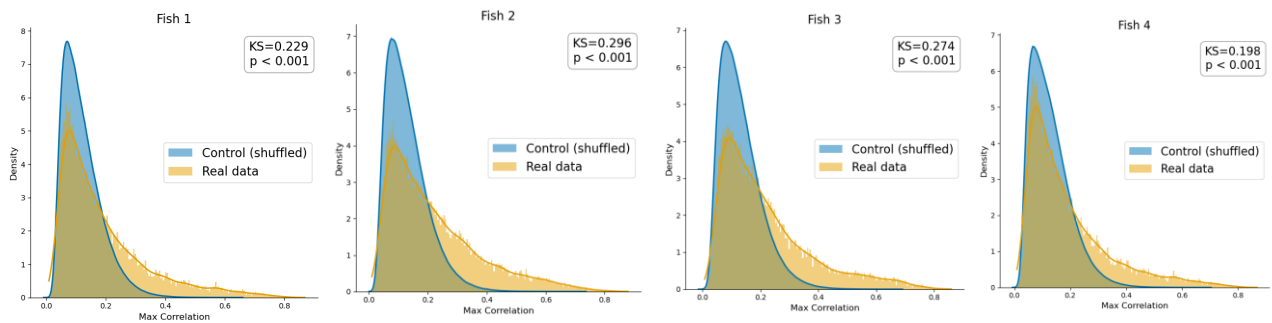


Figure F.2. Gad1b fish had less segmented neurons than vglut2a fish. Boxplot of the number of segmented neurons per fish, for gad1b and vglut2a fish. Individual points correspond to single fish.

Vglut2a fish



Gad1b fish

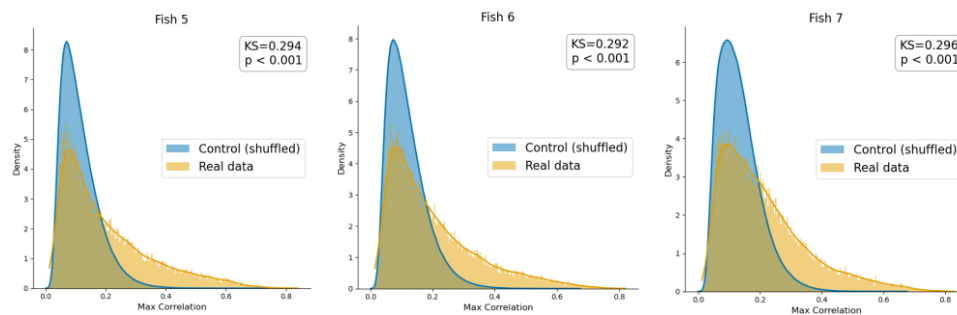


Figure F.3. Observed correlations are significant (data for all fish). Density distributions of maximum correlation value per neuron for control (blue) and real (yellow) data for all fish. (Same plot as in Figure 4.4A).

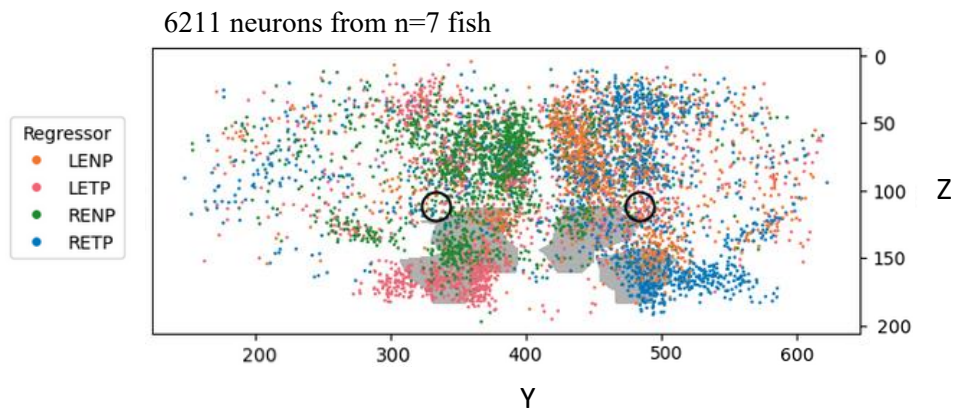


Figure F.4. Nasal-temporal position distinction. Eye position correlation map similar to the ones shown in figures 4.5 but showing a coronal view and both the left and right eye position regressors. Notice how nasal and temporal tuned neurons form distinct groups. Note also that the colours don't match with those from figure 4.5.

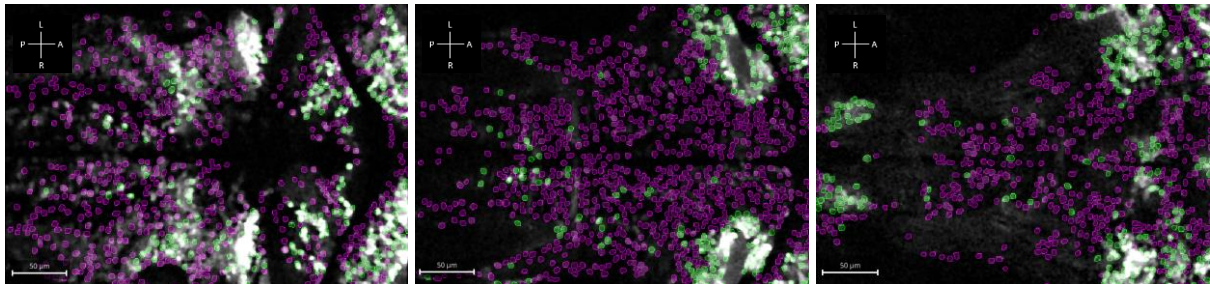


Figure F.5. Classification of neurons from 2-photon imaging stacks as DsRed+ and DsRed-. Each image is an example slice from the red channel of the anatomical stack of one example *vglut2a* fish. Magenta are cells labelled as negative and green are cells labelled as positive.